

Pseudotraces on Almost Unital and Finite-Dimensional Algebras

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Abstract

We introduce the notion of almost unital and finite-dimensional (AUF) algebras, which are associative $\mathbb{C}$ -algebras that may be non-unital or infinite-dimensional, but have sufficiently many idempotents. We show that the pseudotrace construction, originally introduced by Hattori and Stallings for unital finite-dimensional algebras, can be generalized to AUF algebras.

Let A be an AUF algebra. Suppose that G is a projective generator in the category $\text{Coh}_L(A)$ of finitely generated left A -modules that are quotients of free left A -modules, and let $B = \text{End}_{A,-}(G)^{\text{op}}$. We prove that the pseudotrace construction yields an isomorphism between the spaces of symmetric linear functionals $\text{SLF}(A) \xrightarrow{\cong} \text{SLF}(B)$, and that the non-degeneracies on the two sides are equivalent.

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0 Introduction

In [Miy04], Miyamoto introduced the pseudo- q -trace construction for modules of vertex operator algebras (VOAs), generalizing the usual q -trace. His primary motivation was to address the failure of modular invariance for q -traces in the case of C_2 -cofinite but irrational VOAs. While Zhu's theorem in [Zhu96] establishes modular invariance for q -traces in the rational setting, this result does not extend to the irrational case—unless q -traces are replaced with pseudo- q -traces.

Miyamoto's original approach is quite involved. Moreover, his dimension formula for the space of torus conformal blocks is expressed in terms of higher Zhu algebras. This presents two drawbacks: first, higher Zhu algebras are difficult to compute in practice; second, their connection to the VOA module category is not transparent.

Later, Arike [Ari10] and Arike-Nagatomo [AN13] introduced a simplified version of the pseudo- q -trace construction based on the idea of Hattori [Hat65] and Stallings [Sta65]. Below, we briefly outline this approach.

Let A be an algebra, and let B be a unital finite-dimensional algebra. Let M be a finite-dimensional A - B bimodule, projective as a right B -module. By the projectivity, there is a (finite) left coordinate system of M , namely, elements $\alpha_1, \dots, \alpha_n \in \text{Hom}_B(B, M)$ and $\check{\alpha}^1, \dots, \check{\alpha}^n \in \text{Hom}_M(M, B)$ satisfying $\sum_i \alpha_i \circ \check{\alpha}^i = \text{id}_M$. Then the linear map

$$A \rightarrow B \quad x \mapsto \sum_i \check{\alpha}^i \circ x \circ \alpha_i(1_B)$$

descends to a linear map $A/[A, A] \rightarrow B/[B, B]$ which is independent of the choice of the left coordinate system. Its pullback gives a linear map

$$\text{SLF}(B) \rightarrow \text{SLF}(A) \quad \phi \mapsto \text{Tr}^\phi \tag{0.1}$$

where $\text{SLF}(A)$ is the space of symmetric linear functionals on A —that is, linear maps $\psi : A \rightarrow \mathbb{C}$ satisfying $\psi(xy) = \psi(yx)$ for all $x, y \in A$ —and $\text{SLF}(B)$ is the space of symmetric linear functionals on B . The above map is called the **pseudotrace construction**. Note that a typical choice of A is $\text{End}_B(M)$.

The pseudotrace construction is applied to the VOA setting as follows. Let $\mathbb{V}$ be an $\mathbb{N}$ -graded C_2 -cofinite VOA with central charge c , and let $\mathbb{M}$ be a grading-restricted generalized $\mathbb{V}$ -module. Then $\mathbb{M}$ admits a decomposition $\mathbb{M} = \bigoplus_{\lambda \in \mathbb{C}} \mathbb{M}_{[\lambda]}$ into generalized eigenspaces of $L(0)$, where each $\mathbb{M}_{[\lambda]}$ is finite-dimensional. Let $\text{End}_{\mathbb{V}}(\mathbb{M})$ be the algebra of linear operators on $\mathbb{M}$ commuting with the action of $\mathbb{V}$, which is necessarily unital and finite-dimensional. Let B be a unital subalgebra of $\text{End}_{\mathbb{V}}(\mathbb{M})^{\text{op}}$. Assume that $\mathbb{M}$ is a projective right B -module, equivalently, each $\mathbb{M}_{[\lambda]}$ is B -projective. Let $\phi \in \text{SLF}(B)$. Then for $v \in \mathbb{V}$, the expression

$$\text{Tr}^\phi(Y_{\mathbb{M}}(v, z)q^{L(0) - \frac{c}{24}}) = \sum_{\lambda \in \mathbb{C}} \text{Tr}^\phi(P(\lambda)Y_{\mathbb{M}}(v, z)q^{L(0) - \frac{c}{24}}P(\lambda)) \tag{0.2}$$

converges absolutely for $z \in \mathbb{C}$ and $0 < |q| < 1$, and defines a torus conformal block. Here, $P(\lambda)$ is the projection of $\overline{\mathbb{M}} := \prod_{\mu \in \mathbb{C}} \mathbb{M}_{[\mu]}$ onto $\mathbb{M}_{[\mu]}$. Then each $P(\lambda)Y_{\mathbb{M}}(v, z)q^{L(0) - \frac{c}{24}}P(\lambda)$

is a linear operator on $\mathbb{M}_{[\lambda]}$ commuting with the right action of B , and hence Tr^ϕ can be defined on it.

Based on this formulation, in [GR19, Conjecture 5.8], Gainutdinov and Runkel proposed a conjecture that directly relates the space of torus conformal blocks of a C_2 -cofinite VOA $\mathbb{V}$ to the linear structure of the category $\mathrm{Mod}(\mathbb{V})$ of grading-restricted generalized $\mathbb{V}$ -modules. Let $\mathbb{G}$ be a projective generator in $\mathrm{Mod}(\mathbb{V})$, and let $B = \mathrm{End}_{\mathbb{V}}(\mathbb{G})$. Then $\mathbb{G}$ is B -projective. The conjecture asserts that the linear map sending each $\phi \in \mathrm{SLF}(B)$ to (0.2) defines an isomorphism between $\mathrm{SLF}(B)$ and the space of torus conformal blocks of $\mathbb{V}$.

The purpose of this note is to establish results in the theory of associative algebras that are essential for proving the Gainutdinov-Runkel conjecture. The actual resolution of the conjecture will appear in the forthcoming paper [GZ25].

Our approach stems from recognizing a structural analogy between the Gainutdinov-Runkel conjecture and a classical result in associative algebra: If A is a unital finite-dimensional algebra and M is a projective generator in the category of finite-dimensional left A -modules, then M is projective over $B := \mathrm{End}_A(M)^{\mathrm{op}}$, and the pseudotracer map (0.1) is a linear isomorphism. This result was suggested in [BBG21, Sec. 2] and was proved in [Ari10] in the special case that $M = Ae$ where e is a basic idempotent.

However, this classical result is not directly applicable to the Gainutdinov-Runkel conjecture. We need to generalize it to a larger class of associative algebras than unital finite-dimensional ones. In particular, we must consider infinite-dimensional algebras that can be approximated, in a certain sense, by finite-dimensional (and possibly unital) algebras. The need to consider infinite-dimensional associative algebras in the study of irrational VOAs has also been recognized in recent years from different perspectives, such as Huang's associative algebra $A^\infty(\mathbb{V})$ introduced in [Hua24], and the mode transition algebra introduced by Damiolini-Gibney-Krashen in [DGK25].

The infinite-dimensional algebra required for the proof of the Gainutdinov-Runkel conjecture is different from the above mentioned algebras. In [GZ25], we will show that the end

$$\mathbb{E} := \int_{M \in \mathrm{Mod}(\mathbb{V})} M \otimes_{\mathbb{C}} M'$$

a priori an object of $\mathrm{Mod}(\mathbb{V}^{\otimes 2})$, carries a structure of an associative $\mathbb{C}$ -algebra that is compatible with its $\mathbb{V}^{\otimes 2}$ -module structure. This algebra $\mathbb{E}$ is an example of an **almost unital and finite-dimensional algebra**¹ (abbreviated as **AUF algebra**), meaning that $\mathbb{E}$ has a collection of mutually orthogonal idempotents $(e_i)_{i \in \mathcal{I}}$ such that $\mathbb{E} = \sum_{i,j \in \mathcal{I}} e_i \mathbb{E} e_j$ where each summand $e_i \mathbb{E} e_j$ is finite-dimensional. (This sum is automatically direct.) In fact, $\mathbb{E}$ has only finitely many irreducibles. We call such algebra **strongly AUF**.

The main result of this note is a generalization of the aforementioned isomorphism between spaces of symmetric linear functionals to the setting of strongly AUF algebras. More precisely, we prove that the pseudotracer construction defines a linear isomorphism $\mathrm{SLF}(B) \simeq \mathrm{SLF}(A)$ where A is strongly AUF, M is a projective generator of the category $\mathrm{Coh}_L(A)$ of **coherent left A -modules** (i.e., finitely generated left A -modules that are quotients of free ones), and $B = \mathrm{End}_A(M)^{\mathrm{op}}$. See Thm. 9.4. Moreover, we show that

¹Here, “almost” modifies the entire phrase “unital and finite-dimensional”, not just “unital”.

the symmetric linear functional on B is non-degenerate if and only if the corresponding functional on A is non-degenerate. See Thm. 10.4.

Since the associative algebra structure on the end $\mathbb{E}$ will not be developed in this note, we present some alternative examples of AUF algebras for illustration. Let $U(\mathbb{V})$ be the universal algebra of $\mathbb{V}$ as defined in [FZ92]. Let

$$U(\mathbb{V})^{\text{reg}} = \bigoplus_{\lambda, \mu \in \mathbb{C}} U(\mathbb{V})_{[\lambda, \mu]}$$

where $U(\mathbb{V})_{[\lambda, \mu]}$ is the subspace of joint generalized-eigenvectors of the left and right actions of $L(0)$ corresponding to the eigenvalues λ and μ respectively. The following properties are shown in [MNT10]: Each $U(\mathbb{V})_{[\lambda, \mu]}$ is finite-dimensional. For each $\lambda, \mu, \nu \in \mathbb{C}$ one has

$$U(\mathbb{V})_{[\lambda, \mu]} U(\mathbb{V})_{[\mu, \nu]} \subset U(\mathbb{V})_{[\lambda, \nu]}$$

In particular, $U(\mathbb{V})^{\text{reg}}$ is a subalgebra of $U(\mathbb{V})$. Moreover, there is an increasing sequence of idempotents $(1_n)_{n \in \mathbb{Z}_+}$ such that $U(\mathbb{V})^{\text{reg}} = \bigcup_n 1_n U(\mathbb{V})^{\text{reg}} 1_n$. (See [MNT10, Sec. 2.6].) Therefore, $U(\mathbb{V})^{\text{reg}}$ is AUF, since the family of orthogonal idempotents in the definition of AUF algebras can be chosen to be $(1_{n+1} - 1_n)_{n \in \mathbb{Z}_+}$.

For a more elementary and concrete example, consider the following. Let B be a unital finite-dimensional algebra. Let M be a right B -modules. Equip M with a grading

$$M = \bigoplus_{i \in \mathfrak{I}} M(i)$$

where each $M(i)$ is finite-dimensional and is preserved by the right action of B . Let A be

$$\begin{aligned} \text{End}_B^0(M) &:= \{T \in \text{End}(M) : (Tm)b = T(mb) \text{ for all } m \in M, b \in B, \\ &\quad T|_{M(i)} = 0 \text{ for all but finitely many } i \in \mathfrak{I}\} \end{aligned}$$

Then A is clearly an AUF algebra, with the family of mutually orthogonal idempotents given by the projections e_i of M onto $M(i)$.

In fact, any strongly AUF algebra arises from such a construction. More precisely, an algebra is strongly AUF if and only if it is isomorphic to some $\text{End}_B^0(M)$, where M and B satisfy the above conditions and, in addition, M is a projective generator in the category of right B -modules. See Thm. 11.9.

Note that the relationship between $\text{End}_B^0(M)$ and C_2 -cofinite VOAs is straightforward: If $\mathbb{M} \in \text{Mod}(\mathbb{V})$ is equipped with the grading $\bigoplus_{\lambda \in \mathbb{C}} \mathbb{M}_{[\lambda]}$ given by the generalized eigenspaces of $L(0)$, and if B is a unital subalgebra of $\text{End}_{\mathbb{V}}(\mathbb{M})^{\text{op}}$ such that $\mathbb{M}$ is projective as a right B -module, then each $P(\lambda)Y_{\mathbb{M}}(v, z)q^{L(0) - \frac{c}{24}}P(\lambda)$ appearing in (0.2) lies in $\text{End}_B^0(\mathbb{M})$. Therefore, the main result of this note on pseudotraces (Thm. 10.4) can be applied to C_2 -cofinite VOAs. Details of this application will be presented in [GZ25].

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1 Preliminaries

Throughout this note, algebras are associative, not necessarily unital, and over $\mathbb{C}$. Let $\mathbb{N} = \{0, 1, 2, \dots\}$ and $\mathbb{Z}_+ = \{1, 2, \dots\}$. For any vector spaces V, W , we let $\text{Hom}(V, W) = \text{Hom}_{\mathbb{C}}(V, W)$ be the space of linear maps $V \rightarrow W$, and let $\text{End}(V) = \text{Hom}(V, V)$.

Let A be an algebra. Its opposite algebra is denoted by A^{op} . If M, N are left (resp. right) A -modules, we let $\text{Hom}_{A,-}(M, N)$ (resp. $\text{Hom}_{-,A}(M, N)$) be the space of linear maps $M \rightarrow N$ intertwining the left (resp. right) actions of A .

An **idempotent** $e \in A$ is an element satisfying $e^2 = e$. If $e, f \in A$ are idempotent, we write $e \leq f$ if $ef = fe = e$. Equivalently, $f = e + e'$ where $e' \in A$ is an idempotent **orthogonal** to e (i.e. $ee' = e'e = 0$). We say that a nonzero idempotent e is **primitive** if the only idempotent f satisfying $f \leq e$ is $f = 0$ and $f = e$.

In this section, we review some well-known facts about associative algebras. Since, unlike many references, our algebras are not assumed to be unital, we include proofs for the reader's convenience.

Definition 1.1. Let $u, v \in A$. We say that (u, v) is pair of **partial isometries** in A if the following are true:

- (a) $p := vu$ and $q := uv$ are idempotents.
- (b) $u \in qAp$ and $v \in pAq$.

In this case, we also say that u is a partial isometry from p to q , and that v is a partial isometry from q to p . We say that two idempotents are **equivalent** if there are partial isometries between them.

Proposition 1.2. Let $e, f \in A$ be idempotents. Then an element of $\text{Hom}_{A,-}(Ae, Af)$ is precisely the right multiplication of an element of eAf . In particular, we have an algebra isomorphism

$$\text{End}_{A,-}(Ae)^{\text{op}} \simeq eAe$$

Proof. Clearly the right multiplication by some element of eAf yields an element of $\text{Hom}_{A,-}(Ae, Af)$. Conversely, suppose that $T \in \text{Hom}_{A,-}(Ae, Af)$. Let $x = T(e)$, which belongs to Af . Since $ex = eT(e) = T(ee) = T(e) = x$, we see that $x \in eAf$. For each $y \in A$, we have $T(ye) = yT(e) = yx = yex$, which shows that T is the right multiplication by x . $\square$

Corollary 1.3. Let e, f be idempotents in A . The following are equivalent:

- (1) $Ae \simeq Af$ as left A -modules.
- (2) There is a partial isometry from e to f .

Proof. (1) $\Rightarrow$ (2): Let $T \in \text{Hom}_{A,-}(Ae, Af)$ be an isomorphism with inverse $T^{-1} \in \text{Hom}_{A,e}(Af, Ae)$. By Prop. 1.2, T and T^{-1} are realized by the right multiplications of $u \in eAf$ and $v \in fAe$ respectively. Since $TT^{-1} = 1_{Af}$, we have $vu = f$. Since $T^{-1}T = 1_{Ae}$, we have $uv = e$.

(2) $\Rightarrow$ (1): Let $u \in eAf$ and $v \in fAe$ such that $uv = e, vu = f$. Then the right multiplication of u on Ae has inverse being the right multiplication of v . So $Ae \simeq Af$. $\square$

Corollary 1.4. *Let $e \in A$ be an idempotent. Let M be a left A -submodule of Ae . The following are equivalent.*

- (1) M is a direct summand of Ae .
- (2) $M = Af$ for some idempotent $f \leq e$ in A .

Proof. (2) $\Rightarrow$ (1): $Ae = Af \oplus Af'$ where $f' = e - f$ is an idempotent.

(1) $\Rightarrow$ (2): Let $Ae = M \oplus N$. Let $\varphi : Ae \rightarrow Ae$ be the projection on M vanishing on N . Then $\varphi \in \text{End}(Ae)$. By Prop. 1.2, φ is the right multiplication by some $f \in eAe$. Since $\varphi \circ \varphi = \varphi$, clearly $f^2 = f$. Moreover, $M = \varphi(Ae) = (Ae)f = Af$. $\square$

Corollary 1.5. *Let $e \in A$ be an idempotent. The following are equivalent.*

- (1) Ae is an indecomposable left A -module.
- (2) e is primitive.

Proof. This follows immediately from Cor. 1.4. $\square$

Lemma 1.6. *Let M be a nonzero finitely-generated left A -module. Then M has a maximal proper left A -submodule N . Consequently, there is an epimorphism of M onto an irreducible module.*

Proof. Let $\xi_1, \dots, \xi_n$ generate M . Without loss of generality, we assume that ξ_1 does not belong to the submodule N_0 generated by $\xi_2, \dots, \xi_n$. By Zorn's lemma, there is a left submodule $N \leq M$ maximal with respect to the property that $N_0 \subset N$ and $\xi_1 \notin N$. Let us prove that N is a maximal proper submodule. Let $N < K \leq M$. Then by the maximality of N we must have $\xi_1 \in K$. So $\xi_1, \dots, \xi_n \in K$, and hence $K = M$. So K is not proper. $\square$

2 Almost unital algebras

In this section, we introduce the notion of almost unital algebras, which is weaker than being almost unital and finite-dimensional.

Definition 2.1. We say that an algebra A is **almost unital** if the following conditions are satisfied:

- (a) For each $x \in A$, there is an idempotent $e \in A$ such that $x = exe$.
- (b) For any finitely many idempotents $e_1, \dots, e_n \in A$ there exists an idempotent $e \in A$ such that $e_i \leq e$ for all $1 \leq i \leq n$.

Throughout this section, unless otherwise stated, A is assumed to be almost unital.

Definition 2.2. We say that a left A -module M is **quasicoherent** if one of the following equivalent conditions hold:

- (1) For each $\xi \in M$ we have $\xi \in A\xi$.
- (2) For each $\xi \in M$ there exists an idempotent $e \in A$ such that $\xi = e\xi$.

(3) M is a quotient module of $\bigoplus_{i \in I} Ae_i$ where each $e_i \in A$ is an idempotent.

(4) M is a quotient module of a free left A -module $A^{\oplus I}$.

The category of quasicohherent left A -modules is denoted by $\mathbf{QCoh}_L(A)$.

Proof of equivalence. (1) $\Rightarrow$ (2): For each $\xi \in M$, since $\xi \in A\xi$, we have $\xi = a\xi$ for some $a \in A$. Choose idempotent $e \in A$ such that $a \in eAe$. Then $e\xi = ea\xi = a\xi = \xi$.

(2) $\Rightarrow$ (1): Obvious.

(2) $\Rightarrow$ (3): For each $\xi \in M$, let $e_\xi \in A$ be an idempotent such that $e_\xi\xi = \xi$. Then we have a morphism $\bigoplus_{\xi \in M} Ae_\xi \rightarrow M$ whose restriction to Ae_ξ sends each $a \in Ae_\xi$ to $a\xi$. Then $\xi = e_\xi\xi$ implies that $\xi \in Ae_\xi\xi$, and hence ξ is in the range of this morphism. So this morphism is surjective.

(3) $\Rightarrow$ (4): This is obvious, since we have an epimorphism $A \rightarrow Ae_i$ and hence an epimorphism $\bigoplus_{i \in I} A \rightarrow \bigoplus_{i \in I} Ae_i$.

(4) $\Rightarrow$ (2): It suffices to show that $A^{\oplus I}$ satisfies the requirement of (2). Choose $\xi = (a_i)_{i \in I} \in A^{\oplus I}$. Then there are only finitely many $i \in I$ such that $a_i \neq 0$. Since A is almost unital, there exist idempotents $e_i \in A$ (where $i \in I$) such that $a_i = e_i a_i e_i$ for all $i \in I$. (If $a_i = 0$, then we choose $e_i = 0$). Choose idempotent $e \in A$ such that $e_i \leq e$ for all $i \in I$. Then $\xi = e\xi$. $\square$

Definition 2.3. A left A -module M is called **coherent** if it is quasicohherent and finitely-generated. By the above proof of equivalence, it is clear that M is coherent iff M is a quotient of $\bigoplus_{i \in I} Ae_i$ where I is a *finite* index set and $e_i \in A$ is an idempotent. The category of coherent left A modules is denoted by $\mathbf{Coh}_L(A)$.

However, note that a coherent left A -module is not necessarily a quotient of $A^{\oplus n}$ where $n \in \mathbb{Z}_+$. Indeed, A is not necessarily finitely generated as a left A -module.

Remark 2.4. If $M \in \mathbf{QCoh}_L(A)$, then every submodule of M is quasicohherent, and every quotient module of M is quasicohherent M . However, if $M \in \mathbf{Coh}_L(A)$, then a submodule of M is not known to be coherent. Thus, $\mathbf{QCoh}_L(A)$ is an abelian category, while $\mathbf{Coh}_L(A)$ is not known to be abelian.

Proposition 2.5. Let $M \in \mathbf{QCoh}_L(A)$. The following are equivalent.

(1) M is projective in the category of left A -modules.

(2) M is projective in $\mathbf{QCoh}_L(A)$.

(3) M is a direct summand of $\bigoplus_{i \in I} Ae_i$ for some index set I and each $e_i \in A$ is an idempotent.

Proof. (3) $\Rightarrow$ (1): It is well-known that a direct summand of a projective module is projective. Thus, it suffices to prove that $\bigoplus_{i \in I} Ae_i$ is projective. Let $\Phi : \bigoplus_{i \in I} Ae_i \rightarrow N$ be an epimorphism where N is a left A -module. Let $\Gamma : K \rightarrow N$ be an epimorphism. Let

$$\eta_i = \Phi(e_i)$$

Since Γ is surjective, there is $\xi_i \in K$ such that $\Gamma(\xi_i) = \eta_i$. Define $\Psi : \bigoplus_{i \in I} Ae_i \rightarrow K$ to be the morphism sending each $ae_i \in Ae_i$ to $ae_i\xi_i$. Then the following commute:

$$\begin{array}{ccc} & \bigoplus_{i \in I} Ae_i & \\ \Psi \swarrow & \downarrow \Phi & \\ K & \xrightarrow{\Gamma} & N \end{array} \quad \begin{array}{ccc} & ae_i & \\ \swarrow & \downarrow & \\ ae_i\xi_i & \mapsto & ae_i\eta_i \end{array}$$

Note that $\mapsto$ holds since $\Gamma(ae_i\xi_i) = ae_i\Gamma(\xi_i) = ae_i\eta_i$, and $\downarrow$ holds since $\Phi(ae_i) = \Phi(ae_ie_i) = ae_i\Phi(e_i) = ae_i\eta_i$.

(1) $\Rightarrow$ (2): Obvious.

(2) $\Rightarrow$ (3): Choose an epimorphism $\bigoplus_{i \in I} Ae_i \rightarrow M$, which splits because M is projective. So M is a direct summand of $\bigoplus_{i \in I} Ae_i$. $\square$

Proposition 2.6. *Let $M \in \text{Coh}_L(A)$. The following are equivalent.*

- (1) M is projective in the category of left A -modules.
- (2) M is projective in $\text{QCoh}_L(A)$.
- (3) M is projective in $\text{Coh}_L(A)$.
- (4) M is a direct summand of $\bigoplus_{i \in I} Ae_i$ for some finite index set I and each $e_i \in A$ is an idempotent.

Therefore, there is no ambiguity when talking about projective coherent left A -modules.

Proof. Clearly we have (1) $\Rightarrow$ (2) and (2) $\Rightarrow$ (3). By Prop. 2.5 we have (4) $\Rightarrow$ (1). Assume (3). By Rem. 2.4, there is an epimorphism $\bigoplus_{i \in I} Ae_i \rightarrow M$ such that I is finite, and that it splits (because M is projective in $\text{Coh}_L(A)$). So (4) is true. $\square$

Remark 2.7. If $M \in \text{QCoh}_L(A)$, clearly M is irreducible in $\text{QCoh}_L(A)$ iff M is irreducible in the category of left A -modules; in this case we say that M is **irreducible**. Note that even if $M \in \text{Coh}_L(A)$, its irreducibility is understood as in $\text{QCoh}_L(A)$ but not as in $\text{Coh}_L(A)$.

Proposition 2.8. *Let M be a left A -module. The following are equivalent.*

- (1) $M \in \text{QCoh}_L(A)$ and M is irreducible.
- (2) $M \simeq Ae/N$ where $e \in A$ is an idempotent and N is a maximal (proper) left ideal of Ae .
- (3) $M \simeq A/N$ where N is a maximal proper left A -submodule of A .

Proof. (1) $\Rightarrow$ (2): Let $M \in \text{QCoh}_L(A)$ be irreducible. By Def. 2.2, M has an epimorphism Φ from some $\bigoplus_i Ae_i$ where $e_i \in A$ is an idempotent. The restriction of Φ to some Ae_i must be nonzero, and hence must be surjective (since M is irreducible). It follows that M has an epimorphism Ψ from Ae_i . Then $N = \text{Ker}\Psi$ is a maximal proper left A -submodule of Ae_i , and $M \simeq Ae_i/N$.

(1) $\Rightarrow$ (3): In the above proof, M also has an epimorphism from $\bigoplus_i A$ (since we have an epimorphism $A \rightarrow Ae_i$). Thus, replacing Ae_i with A_i in the above proof, we are done.

(2),(3) $\Rightarrow$ (1): Clearly M is irreducible. That $M \in \text{QCoh}_L(A)$ follows from Def. 2.2. $\square$

3 Projective covers

Let A be an algebra, not necessarily almost unital. In this section, we recall some basic facts about projective covers. When A is unital, these results can be found in [AF92], for example. In the non-unital case, one can reduce to the unital setting by considering the unitalization of A . For the reader's convenience, we include complete proofs.

3.1 Basic facts

Definition 3.1. Let M be a left A -module. A left A -submodule $K \leq M$ is called **superfluous**, if for any left A -submodule $L \leq M$ satisfying $K + L = M$ we must have $L = M$.

Remark 3.2. Obviously, we have an equivalent description of superfluous submodules: Let $\pi : M \rightarrow M/K$ be the quotient map. Then $K \leq M$ is superfluous iff for any morphism of left A -modules $\varphi : N \rightarrow M$ such that $\pi \circ \varphi : N \rightarrow M/K$ is surjective, it must be true that φ is surjective.

Definition 3.3. Let M be a left A -module. A **projective cover** of M denotes a left A -module epimorphism $\varphi : P \twoheadrightarrow M$ where P is a projective left A -module, and $\text{Ker} \varphi$ is superfluous in P .

The following property says that among the projective modules that have epimorphisms to M , the projective cover is the smallest one in the sense of direct summand.

Proposition 3.4. Let $\varphi : P \rightarrow M$ be a projective cover of M . Let $\psi : Q \rightarrow M$ be an epimorphism where Q is projective. Then there is a morphism $\alpha : Q \rightarrow P$ such that the following diagram commutes.

$$\begin{array}{ccc} & P & \\ \alpha \nearrow & \downarrow \varphi & \\ Q & \xrightarrow{\psi} & M \end{array} \quad (3.1)$$

Moreover, for any such α , there is a left A -submodule $P' \leq Q$ such that $Q = \ker \alpha \oplus P'$ and that $\alpha|_{P'} : P' \xrightarrow{\sim} P$ is an isomorphism.

By setting $L = \ker \alpha$, it follows that (3.1) is equivalent to

$$\begin{array}{ccc} & P & \\ 0 \oplus \text{id}_P \nearrow & \downarrow \varphi & \\ L \oplus P & \xrightarrow{0 \oplus \varphi} & M \end{array} \quad (3.2)$$

Proof. The existence of α follows from that Q is projective and that φ is an epimorphism. Moreover, since $\ker \varphi$ is superfluous and $\varphi \circ \alpha$ is surjective, by Rem. 3.2, α is surjective. Therefore, since P is projective, the epimorphism α splits, i.e., there is a morphism $\beta : P \rightarrow Q$ such that $\alpha \circ \beta : P \rightarrow P$ equals id_P . One sees that $P' = \beta(P)$ fulfills the requirement. $\square$

It follows that projective covers are unique up to isomorphisms:

Corollary 3.5. *Let M be a left A -module with projective covers $\varphi : P \rightarrow M$ and $\psi : Q \rightarrow M$. Then there exists an isomorphism $\alpha : Q \rightarrow P$ such that (3.1) commutes.*

Proof. By Prop. 3.4, there exists α such that (3.1) commutes. It remains to show that α is an isomorphism. We assume that (3.1) equals (3.2). Since $0 \oplus \varphi : L \oplus P \rightarrow M$ is a projective cover, $L + \ker(P) = \ker(0 \oplus \varphi)$ is superfluous, and hence L is superfluous. Thus, since $L + P$ equals $Q = L \oplus P$, we must have $Q = P$ and hence $L = 0$. So $\alpha = 0 \oplus \text{id}_P$ is an isomorphism. $\square$

3.2 Projective covers of irreducibles

Proposition 3.6. *Suppose that $\varphi : P \rightarrow M$ is a projective cover of an irreducible left A -module M . Then P is indecomposable.*

Proof. Suppose that $P = P' \oplus P''$. Then one of $\varphi|_{P'}, \varphi|_{P''}$ (say $\varphi|_{P'}$) is nonzero. Since M is irreducible, $\varphi|_{P'} : P' \rightarrow M$ must be surjective. So the map $P' \hookrightarrow P' \oplus P'' \xrightarrow{\varphi} M$ is surjective. Since $\ker \varphi$ is superfluous, by Rem. 3.2, $P' \hookrightarrow P' \oplus P''$ is surjective, and hence $P'' = 0$. $\square$

Theorem 3.7. *Let $e \in A$ be a primitive idempotent satisfying*

$$\dim eAe < +\infty$$

Let K be any proper left A -submodule of Ae . Then K is superfluous. In other words, the quotient map $Ae \rightarrow Ae/K$ is the projective cover of Ae/K .

Proof. Step 1. Let $\varphi : Ae \rightarrow Ae/K$ be the quotient map. Let L be a submodule of Ae . Assume that $N + K = Ae$; in other words, if we let $\iota : N \hookrightarrow Ae$ be the inclusion, then $\varphi \circ \iota : N \rightarrow Ae/K$ is surjective. Our goal is to show that $N = Ae$.

Since Ae is projective and $\varphi \circ \iota$ is surjective, there is a morphism $\alpha : Ae \rightarrow N$ such that $\varphi = \varphi \circ \iota \circ \alpha$. Let $\beta = \iota \circ \alpha$. Then the following diagram commutes:

$$\begin{array}{ccccc} & & Ae & & \\ & \nearrow \alpha & \downarrow \varphi & \searrow \beta & \\ N & \xleftarrow{\iota} & Ae & \xrightarrow{\varphi} & Ae/K \end{array} \quad (3.3)$$

To prove that ι is surjective, it suffices to show that β is surjective.

Step 2. Suppose that β is not surjective. Let us find a contradiction. Since $\beta \in \text{End}_{A,-}(Ae)$, by Prop. 1.2, β is the right multiplication by some $x \in eAe$. Let $R_x : eAe \rightarrow eAe$ be the right multiplication of x on eAe . Then R_x is not surjective. Otherwise, there exists $a \in A$ such that $R_x(eae) = e$, i.e., $eaex = e$. Then for each $b \in A$, we have $be = beaex = \beta(beae)$, contradicting the fact that β is not surjective.

It is well-known that if T is a linear operator on a finite-dimensional $\mathbb{C}$ -vector space W , then W is the direct sum of generalized eigenspaces of T , and the projection operator of W onto each generalized eigenspace is a polynomial of T . Therefore, R_x has only one

eigenvalue. Otherwise, there is a polynomial p such that $p(R_x) = R_{p(x)}$ is the projection of eAe onto a proper subspace, and hence $p(x)$ is an idempotent in eAe not equal to 0 or e . This is impossible, since e is assumed to be primitive.

Therefore, R_x has a unique eigenvalue, which must be 0 since R_x is not surjective. By linear algebra, R_x is nilpotent. Since $R_{x^n} = (R_x)^n$, it follows that x is nilpotent, and hence β is nilpotent. By (3.3), we have $\varphi = \varphi \circ \beta$, and hence $\varphi = \varphi \circ \beta = \varphi \circ \beta^2 = \varphi \circ \beta^3 = \dots = 0$. This contradicts the fact that φ is a surjection onto a nonzero module, finishing the proof. $\square$

Corollary 3.8. *Let $e \in A$ be a primitive idempotent satisfying $\dim eAe < +\infty$. Then Ae has a unique proper maximal left A -submodule, denoted by $\text{rad}(Ae)$.*

It follows from Thm. 3.7 that Ae is the projective cover of the irreducible $Ae/\text{rad}(Ae)$.

Proof. By Lem. 1.6, Ae has at least one proper maximal left A -submodule. Suppose that $K \neq L$ are proper maximal left A -submodules of M . By the maximality, we have $K + L = M$. By Thm. 3.7, L is superfluous. So $K = M$, impossible. $\square$

4 Left pseudotraces

Let A, B be algebras such that B is unital. Fix an A - B bimodule M . We do not assume that M_B is unital, i.e., $1_B \in B$ acts as the identity on M .

Definition 4.1. A **left coordinate system** of M denotes a collection of morphisms

$$\alpha_i \in \text{Hom}_{-,B}(B, M) \quad \check{\alpha}^i \in \text{Hom}_{-,B}(M, B) \quad (4.1)$$

where i runs through an index set I such that the following conditions hold:

- (a) For each $\xi \in M$, we have $\check{\alpha}^i(\xi) = 0$ for all but finitely many $i \in I$, and $\sum_{i \in I} \alpha_i \circ \check{\alpha}^i(\xi) = \xi$.
- (b) For each $x \in A$ (viewed as an element of $\text{End}_{-,B}(M)$), we have $x \circ \alpha_i = 0$ and $\check{\alpha}^i \circ x = 0$ for all but finitely many $i \in I$.

Remark 4.2. M is a projective right B -module iff there exists $(\alpha_i, \check{\alpha}^i)_{i \in I}$ of the form (4.1) satisfying condition (a).

Proof. Suppose that there exists $(\alpha_i, \check{\alpha}^i)_{i \in I}$ such that (a) holds. Define morphisms of right B -modules

$$\begin{aligned} \Phi : B^{\oplus I} &\rightarrow M & \oplus_i b_i &\mapsto \sum_i \alpha_i(b_i) \\ \Psi : M &\rightarrow B^{\oplus I} & \xi &\mapsto \oplus_i \check{\alpha}^i(\xi) \end{aligned}$$

Then (a) implies that $\Phi \circ \Psi = \text{id}_M$. Thus, M is a direct summand of $B^{\oplus I}$, and hence is projective as a right B -module.

Conversely, assume M is projective as a right B -module. Then we have an epimorphism $\Phi : B^{\oplus I} \rightarrow M$ and a morphism $\Psi : M \rightarrow B^{\oplus I}$ such that $\Phi \circ \Psi = \text{id}_M$. For each

$i \in I$, let $\iota_i : B \rightarrow B^{\oplus I}$ be the inclusion map of B into the i -th direct summand, and $\pi_i : B^{\oplus I} \rightarrow B$ be the projection map onto the i -th direct summand. Set

$$\alpha_i = \Phi \circ \iota_i \quad \check{\alpha}^i = \pi_i \circ \Psi$$

Then $(\alpha_i, \check{\alpha}^i)_{i \in I}$ satisfies (a). □

Definition 4.3. Assume that M has a left coordinate system $(\alpha_i, \check{\alpha}^i)_{i \in I}$. Define the **B -trace function**

$$\mathrm{Tr}^B : A \rightarrow B/[B, B] \quad x \mapsto \sum_{i \in I} \check{\alpha}^i \circ x \circ \alpha_i$$

where the RHS, originally an element of $\mathrm{End}_{-,B}(B) \simeq B$,² is descended to $B/[B, B]$.

Lemma 4.4. The definition of Tr^B is independent of the choice of left coordinate systems.

Proof. Suppose that $(\beta_j, \check{\beta}^j)_{j \in J}$ is another left coordinate system of the A - B bimodule M . Let $I_x \subset I$ and $J_x \subset J$ be finite sets such that $\check{\alpha}^i \circ x = 0, x \circ \alpha_i = 0$ for any $i \in I \setminus I_x$, and that $\check{\beta}^j \circ x = 0, x \circ \beta_j = 0$ for any $j \in J \setminus J_x$. Then

$$\sum_{i \in I_x} \check{\alpha}^i \circ x \circ \alpha_i = \sum_{i \in I_x, j \in J} \check{\alpha}^i \circ x \circ \beta_j \circ \check{\beta}^j \circ \alpha_i = \sum_{i \in I_x, j \in J_x} \check{\alpha}^i \circ x \circ \beta_j \circ \check{\beta}^j \circ \alpha_i$$

Since each $\check{\alpha}^i \circ x \circ \beta_j$ and $\check{\beta}^j \circ \alpha_i$ are in $\mathrm{End}_{-,B}(B) \simeq B$, the RHS above equals

$$\sum_{i \in I_x, j \in J_x} \check{\beta}^j \circ \alpha_i \circ \check{\alpha}^i \circ x \circ \beta_j = \sum_{j \in J_x} \check{\beta}^j \circ x \circ \beta_j$$

in $B/[B, B]$. □

Proposition 4.5. Tr^B is **symmetric**, i.e., $\mathrm{Tr}^B(xy) = \mathrm{Tr}^B(yx)$ for any $x, y \in A$. Therefore, Tr^B descends to a linear map $A/[A, A] \rightarrow B/[B, B]$.

Proof. Let $x, y \in A$. Let $I_0 \subset I$ be a finite set such that $\check{\alpha}^i \circ x = \check{\alpha}^i \circ y = 0$ and $x \circ \alpha_i = y \circ \alpha_i = 0$ for all $i \in I \setminus I_0$. Then

$$\mathrm{Tr}^B(xy) = \sum_{i \in I_0} \check{\alpha}^i \circ x \circ y \circ \alpha_i = \sum_{i, j \in I_0} \check{\alpha}^i \circ x \circ \alpha_j \circ \check{\alpha}^j \circ y \circ \alpha_i$$

and similarly

$$\mathrm{Tr}^B(yx) = \sum_{i, j \in I_0} \check{\alpha}^j \circ y \circ \alpha_i \circ \check{\alpha}^i \circ x \circ \alpha_j$$

The two RHS's are equal in $B/[B, B]$, noting that $\check{\alpha}^i \circ x \circ \alpha_j$ and $\check{\alpha}^j \circ y \circ \alpha_i$ are both in $\mathrm{End}_{-,B}(B) \simeq B$. □

²This isomorphism relies on the fact that B is unital.

Definition 4.6. Let $\phi : B \rightarrow \mathbb{C}$ be a **symmetric linear functional (SLF)**, i.e., a linear map satisfying $\phi(ab) = \phi(ba)$ for all $a, b \in B$. The (left) **pseudotrace** associated to ϕ (and M), denoted by Tr^ϕ , is defined to be

$$\text{Tr}^\phi = \phi \circ \text{Tr}^B : A \rightarrow \mathbb{C} \quad (4.2)$$

It is an SLF on A .

Thus, for each $x \in A$ we have

$$\text{Tr}^\phi(x) = \sum_{i \in I} \phi(\tilde{\alpha}^i \circ x \circ \alpha_i(1_B)) \quad (4.3)$$

5 AUF algebras and projective covers of irreducibles

Definition 5.1. An algebra A is called **almost unital and finite-dimensional (AUF)** if there is a family of mutually orthogonal idempotents $(e_i)_{i \in \mathfrak{I}}$ such that the following conditions hold:

- (a) For each $i, j \in \mathfrak{I}$ we have $\dim e_i A e_j < +\infty$.
- (b) $A = \sum_{i, j \in \mathfrak{I}} e_i A e_j$. (That is, for each $x \in A$ one can find a finite subset $I \subset \mathfrak{I}$ and a collection $(x_{i, j})_{i, j \in I}$ such that $x = \sum_{i, j \in I} e_i x_{i, j} e_j$.)

Note that (b) automatically implies $A = \bigoplus_{i, j \in \mathfrak{I}} e_i A e_j$.

It is illuminating to view an element $x \in A$ as an $\mathfrak{I} \times \mathfrak{I}$ matrix whose (i, j) -entry is $e_i x e_j$.

Remark 5.2. Each AUF algebra A is almost unital.

Proof. For each $x_1, \dots, x_n \in A$, we can find a subset $I_0 \subset \mathfrak{I}$ such that $x \in e' A e'$, where $e' = \sum_{i \in I_0} e_i$. By choose $n = 1$ and $x_1 = x \in A$, we see $x = e' x e'$. By choosing idempotents $x_i = e_i \in A$, we see $e_i \leq e'$ for all $1 \leq i \leq n$. $\square$

Lemma 5.3. In Def. 5.1, one can assume moreover that each e_i is primitive (in A).

Proof. Let $(e_i)_{i \in \mathfrak{I}}$ be as in Def. 5.1. For each $i \in \mathfrak{I}$, since $e_i A e_i$ is a finite-dimensional left $e_i A e_i$ -module, it is a finite direct sum of indecomposable left $e_i A e_i$ -submodules. By Cor. 1.4 and 1.5, we have a finite direct sum $e_i A e_i = \bigoplus_{k \in \mathfrak{K}_i} e_i A f_{i, k}$ where $(f_{i, k})_{k \in \mathfrak{K}_i}$ is a finite family of mutually orthogonal idempotents in $e_i A e_i$, that $\sum_k f_{i, k} = e_i$, and that each $f_{i, k}$ is primitive in $e_i A e_i$. Clearly $f_{i, k}$ is also primitive in A . Replacing $(e_i)_{i \in \mathfrak{I}}$ by $(f_{i, k})_{i \in \mathfrak{I}, k \in \mathfrak{K}_i}$ does the job. $\square$

In the remaining part of this section, we always assume that A is AUF.

Remark 5.4. For each idempotents $e, f \in A$, we have

$$\dim e A f < +\infty$$

Indeed, one can find a finite set $I_0 \subset \mathcal{I}$ such that $e, f \in e' Ae'$ where $e' = \sum_{i \in I_0} e_i$. Then $\dim e' Ae' < +\infty$, and hence $\dim eAf < +\infty$.

It follows that each idempotent $e \in A$ has a (finite) orthogonal primitive decomposition $e = \varepsilon_1 + \cdots + \varepsilon_n$. This follows from a decomposition of the finite-dimensional left eAe -module eAe into indecomposable submodules. $\square$

Recall Rem. 2.7 about irreducibility.

Theorem 5.5. *The following are true.*

1. For each primitive idempotent $e \in A$, let $\text{rad}(Ae)$ be the unique proper maximal left submodule of Ae (cf. Cor. 3.8). Then $Ae \rightarrow Ae/\text{rad}(Ae)$ gives a projective cover of the irreducible coherent module $Ae/\text{rad}(Ae)$.
2. Any irreducible $M \in \text{QCoh}_L(A)$ is isomorphic to $Ae/\text{rad}(Ae)$ for some primitive idempotent $e \in A$.
3. Let e, f be primitive idempotents. Then the following are equivalent:
 - (1) $Ae \simeq Af$ as left A -modules.
 - (2) $Ae/\text{rad}(Ae) \simeq Af/\text{rad}(Af)$ as left A -modules.
 - (3) $e \simeq f$, i.e., there is a partial isometry (in A) from e to f .

Proof. Part 1 was already proved, cf. Thm. 3.7. (Note that Thm. 3.7 and its consequences are applicable since $\dim eAe < +\infty$ by Rem. 5.4.)

Part 2: By Prop. 2.8, M has an epimorphism Ψ from A . Let $(e_i)_{i \in \mathcal{I}}$ be as in Def. 5.1 such that each e_i is primitive (Lem. 5.3). Then $A \simeq \bigoplus_i Ae_i$ as left A -modules. The restriction of Ψ to some Ae_i must be nonzero, and hence must be surjective. Therefore $M \simeq Ae_i/\text{rad}(Ae_i)$.

Part 3: (1) $\Rightarrow$ (2) is obvious. (2) $\Rightarrow$ (1) follows from the uniqueness of projective covers (Cor. 3.5). (1) $\Leftrightarrow$ (3) follows from Cor. 1.3. $\square$

Corollary 5.6. *Let $P \in \text{Coh}_L(A)$. The following are equivalent.*

- (1) P is projective and indecomposable.
- (2) P is the projective cover of an irreducible $M \in \text{QCoh}_L(A)$, which (by Thm. 5.5) is isomorphic to Ae for some primitive idempotent $e \in A$.

Proof. (2) $\Rightarrow$ (1): This follows from Prop. 3.6.

(1) $\Rightarrow$ (2): By Lem. 1.6, P has an epimorphism to an irreducible, which (by Thm. 5.5) is of the form $Ae/\text{rad}(Ae)$ where $e \in A$ is a primitive idempotent. We know that Ae is its projective cover. Since P is projective, by Prop. 3.4, Ae is a direct summand of P . Since P is indecomposable, we must have $P = Ae$. $\square$

6 Pseudotraces and generating idempotents of strongly AUF algebras

Let A be AUF. In this section, we show that if $e \in A$ is a generating idempotent, any SLF ψ on A can be recovered from $\psi|_{eAe}$ via the pseudotrace construction.

Definition 6.1. An idempotent $e \in A$ is called **generating** if every irreducible $M \in \text{QCoh}_L(A)$ has an epimorphism from Ae .

Proposition 6.2. Let $e \in A$ be an idempotent. Let $e = \varepsilon_1 + \cdots + \varepsilon_n$ be an orthogonal primitive decomposition (cf. Rem. 5.4). The following are equivalent:

- (1) e is generating.
- (2) Any primitive idempotent of A is isomorphic to ε_i for some i .
- (3) Any irreducible $M \in \text{QCoh}_L(A)$ is isomorphic to $A\varepsilon_i/\text{rad}(A\varepsilon_i)$ for some i .

Proof. (1) $\Rightarrow$ (3): Each irreducible $M \in \text{QCoh}_L(A)$ has an epimorphism from $Ae = A\varepsilon_1 \oplus \cdots \oplus A\varepsilon_n$, and hence an epimorphism from some $A\varepsilon_i$. By Cor. 3.8, the kernel of this epimorphism is $\text{rad}(A\varepsilon_i)$. Therefore, we have $A\varepsilon_i/\text{rad}(A\varepsilon_i) \simeq M$.

(3) $\Rightarrow$ (1): Obvious.

(2) $\Leftrightarrow$ (3): Immediate from Thm. 5.5. □

Corollary 6.3. Let $e, f \in A$ be idempotents such that $e \leq f$ and e is a generating idempotent of A . Then e is a generating idempotent of fAf .

Proof. Let p be any primitive idempotent of fAf . Then p is a primitive idempotent of A . By Prop. 6.2, if we let $e = \varepsilon_1 + \cdots + \varepsilon_n$ be an orthogonal primitive decomposition, then there exist $1 \leq i \leq n$ and $u \in \varepsilon_i Ap, v \in pA\varepsilon_i$ such that $uv = \varepsilon_i$ and $vu = p$. So p is isomorphic in fAf to ε_i . By Prop. 6.2, we conclude that e is generating in fAf . □

Corollary 6.4. The following are equivalent.

- (1) A has a generating idempotent.
- (2) $\text{QCoh}_L(A)$ has finitely many equivalence classes of irreducible objects.
- (3) A has finitely many isomorphism classes of primitive idempotents.

If one of these conditions holds, we say that A is **strongly AUF**.

Proof. (1) $\Rightarrow$ (2): Immediate from Prop. 6.2.

(2) $\Leftrightarrow$ (3): Immediate from Thm. 5.5.

(3) $\Rightarrow$ (1): Let $M_1, \dots, M_n \in \text{QCoh}_L(A)$ exhaust all equivalence classes of irreducibles. Let $(e_i)_{i \in \mathcal{J}}$ be as in Def. 5.1. For each $1 \leq k \leq n$, by Prop. 2.8, M_k has an epimorphism from A . Since $A = \bigoplus_{i \in \mathcal{J}} Ae_i$, it follows that M_k has an epimorphism from Ae_{i_k} for some $i_k \in \mathcal{J}$. If we assume at the beginning that $M_1, \dots, M_n$ are mutually non-isomorphic, then $e_{i_1}, \dots, e_{i_n}$ must be distinct, and hence mutually orthogonal. So $e = e_{i_1} + \cdots + e_{i_n}$ is a generating idempotent. □

Theorem 6.5. Assume that A is strongly AUF, and let $e \in A$ be a generating idempotent. Then the A -(eAe) bimodule Ae has a left coordinate system. In particular, by Rem. 4.2, Ae is a projective right eAe -module.

The following construction of left coordinate system is important and is motivated by [Ari10, Lem. 3.9].

Proof. Let $(e_i)_{i \in \mathcal{I}}$ be as in Def. 5.1. By Lem. 5.3, we can assume that each e_i is primitive. Let $e = \varepsilon_1 + \cdots + \varepsilon_n$ be an orthogonal primitive decomposition of e . By Prop. 6.2, there are partial isometries u_i, v_i such that

$$\begin{aligned} v_i u_i &= \varepsilon_{k_i} & u_i v_i &= e_i \\ u_i &\in e_i A \varepsilon_{k_i} & v_i &\in \varepsilon_{k_i} A e_i \end{aligned}$$

where $k_i \in \{1, \dots, n\}$. In particular $u_i \in e_i A e$ and $v_i \in e A e_i$. Let

$$\begin{aligned} \alpha_i &\in \text{End}_{-,eAe}(eAe, Ae) & \check{\alpha}^i &\in \text{End}_{-,eAe}(Ae, eAe) \\ \alpha_i(xe) &= u_i \cdot x e & \check{\alpha}^i(xe) &= v_i \cdot x e \end{aligned}$$

One checks easily that $(\alpha_i, \check{\alpha}^i)_{i \in \mathcal{I}}$ is a left coordinate system. □

The proof of [Ari10, Thm. 3.10] can be easily adapted to prove the following theorem.

Theorem 6.6. Assume that A is strongly AUF, and let $e \in A$ be a generating idempotent. Then there is a linear isomorphism

$$\text{SLF}(A) \xrightarrow{\cong} \text{SLF}(eAe) \quad \psi \mapsto \psi|_{eAe}$$

whose inverse is given by

$$\text{SLF}(eAe) \xrightarrow{\cong} \text{SLF}(A) \quad \phi \mapsto \text{Tr}^\phi$$

Here, Tr^ϕ is the pseudotrace on A with respect to ϕ and the A -(eAe) bimodule Ae .

Proof. Let $u_i, v_i, \alpha_i, \check{\alpha}^i$ be as in the proof of Thm. 6.5. For any $\phi \in \text{SLF}(eAe)$, let us compute Tr^ϕ . Let $x \in A$, viewed as an element of $\text{End}_{-,eAe}(Ae)$. Then $\check{\alpha}^i \circ x \circ \alpha_i \in \text{End}_{-,eAe}(eAe)$ equals (the left multiplication by) $v_i x u_i$. Then

$$\text{Tr}^\phi(x) = \sum_{i \in \mathcal{I}} \phi(v_i x u_i) \tag{6.1}$$

Note that the RHS is a finite sum since $u_i = e_i u_i$, and since $x e_i = 0$ for all but finitely many i .

To show that $\text{Tr}^\phi|_{eAe} = \phi$, we compute

$$\text{Tr}^\phi(xe) = \sum_i \phi(v_i x e u_i) = \sum_i \phi(v_i x e \cdot e u_i)$$

Since $v_i x e, e u_i \in eAe$, and since ϕ is SLF, we have

$$\text{Tr}^\phi(xe) = \sum_i \phi(e u_i \cdot v_i x e) = \sum_i \phi(e e_i x e) = \phi(xe)$$

Finally, let $\psi \in \text{SLF}(A)$. Then for each $x \in A$,

$$\text{Tr}^{\psi|_{eAe}}(x) = \sum_i \psi|_{eAe}(v_i x u_i) = \sum_i \psi(v_i x u_i) = \sum_i \psi(u_i v_i x) = \sum_i \psi(e_i x) = \psi(x)$$

This proves $\text{Tr}^{\psi|_{eAe}} = \psi$. □

7 Projective generators of strongly AUF algebras

Let A be an AUF algebra.

Remark 7.1. A left A -module M is coherent if and only if M is a quotient module of $(Ae)^{\oplus n}$ where $n \in \mathbb{Z}_+$ and $e \in A$ is an idempotent.

Proof. “ $\Leftarrow$ ” is obvious. Conversely, let $M \in \text{Coh}_L(A)$. By Def. 2.3, M is a quotient module of $Ap_1 \oplus \cdots \oplus Ap_n$ where each p_i is an idempotent. By Rem. 5.2, one can find an idempotent $e \in A$ which is $\geq p_1, \dots, p_n$. Then M is a quotient module of $(Ae)^{\oplus n}$. □

Remark 7.2. By Rem. 7.1, if $M \in \text{Coh}_L(A)$ and $x \in A$, then $\dim xM < +\infty$.

Proof. Suppose that M has an epimorphism from $N := (Ae)^{\oplus n}$ where $e \in A$ is an idempotent. Then $\dim xM \leq \dim xN$. Let $f \in A$ be an idempotent such that $x = fxf$. Then $xAe \subset fAe$, and hence

$$\dim xN = n \dim xAe \leq n \dim fAe < +\infty$$

□

7.1 Basic facts

Definition 7.3. Let $\mathcal{S}$ and $\mathcal{T}$ be classes of objects in $\text{Coh}_L(A)$. We say that $\mathcal{S}$ **generates** $\mathcal{T}$ if each object of $\mathcal{T}$ is a quotient of a *finite* direct sum of objects in $\mathcal{S}$.

Definition 7.4. We say that $M \in \text{Coh}_L(A)$ is a **generator** (of $\text{Coh}_L(A)$) if it generates every object of $\text{Coh}_L(A)$, i.e., every $N \in \text{Coh}_L(A)$ is a quotient module of $M^{\oplus n}$ for some $n \in \mathbb{Z}_+$. A generator which is also projective is called a **projective generator**.

Example 7.5. Let $(e_i)_{i \in \mathcal{I}}$ be as in Def. 5.1. Then $\mathcal{S} := \{Ae_i : i \in \mathcal{I}\}$ generates $\text{Coh}_L(A)$.

Proof. By the proof of Rem. 5.2, for any idempotent $e \in A$ one can find a finite set $I_0 \subset \mathcal{I}$ such that $e \leq \sum_{i \in I_0} e_i$. Therefore, $\mathcal{S}$ generates each Ae , and hence (by Rem. 7.1) generates $\text{Coh}_L(A)$. □

Proposition 7.6. Let $M \in \text{Coh}_L(A)$ be projective. The following are equivalent.

- (1) M is a projective generator.
- (2) Each irreducible $N \in \text{Coh}_L(A)$ has an epimorphism from M .

Proof. (1) $\Rightarrow$ (2): Obvious.

(2) $\Rightarrow$ (1): Let $(e_i)_{i \in \mathcal{I}}$ be as in Def. 5.1. By Lem. 5.4, we assume that each e_i is primitive. By Exp. 7.5, it suffices to prove that M generates each Ae_i . By Thm. 5.5, Ae_i is the projective cover of the irreducible $N := Ae_i/\text{rad}(Ae_i)$. By (2), M has an epimorphism to N . Since M is projective, by Prop. 3.4, Ae_i is isomorphic to a direct summand of M . $\square$

Corollary 7.7. *Let $e \in A$ be an idempotent. Then the following are equivalent.*

- (1) Ae is a (necessarily projective) generator.
- (2) e is a generating idempotent.

Proof. (1) $\Rightarrow$ (2): Clear from Def. 6.1. (2) $\Rightarrow$ (1): Immediate from Prop. 7.6. $\square$

Proposition 7.8. *$\text{Coh}_L(A)$ has a projective generator if and only if A is strongly AUF.*

Proof. “ $\Leftarrow$ ” follows from Cor. 6.4 and 7.7. Conversely, if $\text{Coh}_L(A)$ has a projective generator M , by Rem. 7.1, an idempotent $e \in A$ can be found such that Ae generates M , and hence generates $\text{Coh}_L(A)$. So e is a generating idempotent. Thus, by Cor. 6.4, $\text{Coh}_L(A)$ has finitely many irreducibles. So A is strongly AUF. $\square$

7.2 Projective generators and endomorphism algebras

Our next goal is to give criteria for projective generators in terms of the endomorphism algebras. We need the endomorphism algebras to be finite-dimensional:

Proposition 7.9. *Let $M, N \in \text{Coh}_L(A)$. Then*

$$\dim \text{Hom}_{A,-}(M, N) < +\infty$$

Proof. By Def. 2.3, there is an epimorphism from a finite direct sum $\bigoplus_i Ae_i$ to M , where e_i is an idempotent. By taking composition with this epimorphism, we get

$$\text{Hom}_{A,-}(M, N) \rightarrow \text{Hom}_{A,-}\left(\bigoplus_i Ae_i, N\right) \simeq \bigoplus_i \text{Hom}_{A,-}(Ae_i, N) \quad (7.1)$$

where the first map is injective. Thus, it suffices to prove that each $\text{Hom}_{A,-}(Ae_i, N)$ is finite-dimensional.

Again, we can find an epimorphism $\Phi : \bigoplus_j Af_j \twoheadrightarrow N$ (where $\bigoplus_j$ is finite). Since Ae_i is projective, each $\alpha \in \text{Hom}_{A,-}(Ae_i, N)$ can be lifted to some $\beta \in \text{Hom}_{A,-}(Ae_i, \bigoplus_j Af_j)$ such that $\alpha = \Phi \circ \beta$. Thus

$$\dim \text{Hom}_{A,-}(Ae_i, N) \leq \dim \text{Hom}_{A,-}\left(Ae_i, \bigoplus_j Af_j\right) = \sum_j \dim \text{Hom}_{A,-}(Ae_i, Af_j)$$

where $\dim \text{Hom}_{A,-}(Ae_i, Af_j) = \dim e_i Af_j < +\infty$. $\square$

Proposition 7.10. *Let M be a left A -module. Let $B = \text{End}_{A,-}(M)^{\text{op}}$, and let $p, q \in B$ be idempotents. Then an element of $\text{Hom}_{A,-}(Mp, Mq)$ is precisely the right multiplication of an element of pBq . In particular, we have a canonical isomorphism*

$$\text{End}_{A,-}(Mp)^{\text{op}} \simeq pBp$$

Consequently, the direct summands of the left A -module Mp correspond bijectively to the sub-idempotents of p in B .

Proof. This is similar to the proofs of Prop. 1.2 and Cor. 1.4. Any $y \in pBq$ defines a morphism $Mp \rightarrow Mq$ by right multiplication. Conversely, if $T \in \text{Hom}_{A,-}(Mp, Mq)$, let $\hat{T} : M \rightarrow M$ be $\hat{T}(\xi) = T(\xi p)$. Then $\hat{T} \in \text{End}_{A,-}(M)$, and hence $\hat{T}$ is the right multiplication by some $\hat{y} \in B$. Note that $T = \hat{T}|_{Mp}$, and hence $T(\xi p) = \xi p \hat{y}$ for each $\xi \in M$. Since T has range in Mq , we have $T(\xi p) = \xi p \hat{y} q$. So T is the right multiplication by $y := p \hat{y} q \in pBq$. $\square$

Theorem 7.11. *Let $M \in \text{Coh}_L(A)$. Let $B = \text{End}_{A,-}(M)^{\text{op}}$ which is a finite-dimensional unital algebra (by Prop. 7.9). Let $p \in B$ be an idempotent. Consider the following statements:*

- (1) *As coherent left A -modules, Mp generates M .*
- (2) *p is a generating idempotent of B .*

Then (2) $\Rightarrow$ (1). If M is projective, then (1) $\Leftrightarrow$ (2).

Proof. (2) $\Rightarrow$ (1): Since $\dim B < +\infty$, we have a primitive orthogonal decomposition $1_B = q_1 + \cdots + q_n$ where each $q_j \in B$ is a primitive idempotent. By Prop. 6.2, each q_j is isomorphic to a sub-idempotent of p . Thus Mq_j is isomorphic to a direct summand of the left A -module Mp . So Mp generates $\bigoplus_j Mq_j = M$.

(1) $\Rightarrow$ (2): Let q be any primitive idempotent of B . Since Mp generates M and since M generates Mq , we have that Mp generates Mq . We claim that Mq is isomorphic to a direct summand of Mp . Then Prop. 7.10 will imply that q is isomorphic (in B) to a sub-idempotent of p . This implies (2), thanks to Prop. 6.2.

Let us prove the claim, assuming that M is projective. Since Mq is a direct summand of M , we see that Mq is projective. Since q is primitive in B , by Prop. 7.10, Mq is an indecomposable left A -module. Therefore, by Cor. 5.6, Mq is the projective cover of an irreducible $N \in \text{Coh}_L(A)$. Since Mp generates Mq , it generates N . Thus N has an epimorphism from a finite direct sum of Mp . Since N is irreducible, N has an epimorphism from Mp . Note that Mp is also projective. Therefore, by Prop. 3.4, Mq is isomorphic to a direct summand of Mp . $\square$

Corollary 7.12. *Assume that $G \in \text{Coh}_L(A)$ is a projective generator. Let M be a left A -module. Then the following are equivalent.*

- (1) *$M \in \text{Coh}_L(A)$, and M is a projective generator (of $\text{Coh}_L(A)$).*
- (2) *There exist $n \in \mathbb{Z}_+$ and a generating idempotent p of $B := \text{End}_{A,-}(G^{\oplus n})^{\text{op}}$ such that $M \simeq G^{\oplus n} \cdot p$.*

In particular, if $e \in A$ is a generating idempotent, one can take $G = Ae$. Thus a projective generator of $\text{Coh}_L(A)$ is (up to isomorphisms) precisely of the form $(Ae)^{\oplus n}p$ where $n \in \mathbb{Z}_+$ and $p \in \text{End}_{A,-}((Ae)^{\oplus n})^{\text{op}}$ is a generating idempotent.

Proof. (2) $\Rightarrow$ (1): By Thm. 7.11, M generates $G^{\oplus n}$. So M is a generator. Since $G^{\oplus n}p$ is a direct summand of the projective coherent module $G^{\oplus n}$, $G^{\oplus n}p$ is also projective and coherent.

(1) $\Rightarrow$ (2): M has an epimorphism from $G^{\oplus n}$ for some $n \in \mathbb{Z}_+$. Since M is projective, this epimorphism splits. So M can be viewed as a direct summand of $G^{\oplus n}$. Let p be the projection of $G^{\oplus n}$ onto M , which can be viewed as an endomorphism of $G^{\oplus n}$. So p is an idempotent of B , and $M = G^{\oplus n}p$. Since M is a generator, it generates $G^{\oplus n}$. Since $G^{\oplus n}$ is projective, by Thm. 7.11, p is generating. $\square$

8 Right pseudotraces

Let A be an AUF algebra. Let B be a unital algebra. Let M be an A - B bimodule, coherent as a left A -module.

For each $y \in B$ and $\xi \in M$, we write ξy as $y^{\text{op}}\xi$. Namely, y^{op} is viewed as an element of $\text{End}_{A,-}(M)$.

Definition 8.1. A **right coordinate system** of M denotes a collection of morphisms

$$\beta_j \in \text{Hom}_{A,-}(Ae, M) \quad \check{\beta}^j : \text{Hom}_{A,-}(M, Ae)$$

where $e \in A$ is an idempotent (called the **domain idempotent**), and j runs through a *finite* index set J such that the $\sum_{j \in J} \beta_j \circ \check{\beta}^j$ equals id_M .

Remark 8.2. M has a right coordinate system iff M is A -projective.

Proof. By Rem. 7.1, each $N \in \text{Coh}_L(A)$ has an epimorphism from $(Ae)^{\oplus n}$ where $e \in A$ is an idempotent and $n \in \mathbb{Z}_+$. This epimorphism splits iff N is projective in $\text{Coh}_L(A)$. Therefore, similar to Rem. 4.2, we see that M has a right coordinate system iff M is A -projective. $\square$

Remark 8.3. In Def. 8.1, one can freely enlarge the domain idempotent e . More precisely, suppose that $f \in A$ is an idempotent such that $e \leq f$. One can define a new right coordinate system

$$\begin{aligned} \gamma_j \in \text{Hom}_{A,-}(Af, M) & \quad \check{\gamma}^j \in \text{Hom}_{A,-}(M, Af) \\ \gamma_j(af) = \beta_j(ae) & \quad \check{\gamma}^j(\xi) = \check{\beta}^j(\xi) \end{aligned} \tag{8.1}$$

called the **canonical extension** of $(\beta_j, \check{\beta}^j)_{j \in J}$.

Definition 8.4. Assume that M has a right coordinate system $(\beta_j, \check{\beta}^j)_{j \in J}$. For each $\psi \in \text{SLF}(A)$, define the (right) **pseudotrace** ${}^\psi\text{Tr}$ associated to ψ to be

$${}^\psi\text{Tr} : B \rightarrow \mathbb{C} \quad {}^\psi\text{Tr}(y) = \sum_{j \in J} \psi((\check{\beta}^j \circ y^{\text{op}} \circ \beta_j)^{\text{op}})$$

noting that $\check{\beta}^j \circ y^{\text{op}} \circ \beta_j \in \text{End}_{A,-}(Ae) \simeq (eAe)^{\text{op}}$. In other words,

$${}^{\psi}\text{Tr}(y) = \sum_{j \in J} \psi(\check{\beta}^j \circ y^{\text{op}} \circ \beta_j(e)) \quad (8.2)$$

Note that in (8.2) we have $\beta_j(e) \in M$, and hence $\check{\beta}^j \circ y^{\text{op}} \circ \beta_j(e) \in Ae$. So

$$\check{\beta}^j \circ y^{\text{op}} \circ \beta_j(e) = \check{\beta}^j \circ y^{\text{op}} \circ \beta_j(e^2) = e\check{\beta}^j \circ y^{\text{op}} \circ \beta_j(e) \in eAe$$

Proposition 8.5. *Assume that M is A -projective. Let $\psi \in \text{SLF}(A)$. Then ${}^{\psi}\text{Tr} \in \text{SLF}(B)$. Moreover, the definition of ${}^{\psi}\text{Tr}$ is independent of the choice of right coordinate systems.*

Proof. From (8.1) and (8.2), it is clear that a canonical extension of the right coordinate system does not affect the value of ${}^{\psi}\text{Tr}(y)$. Also, note that since A is AUF, for any idempotents $e_1, e_2 \in A$ there is an idempotent e_3 such that $e_1, e_2 \leq e_3$. Therefore, to compare ${}^{\psi}\text{Tr}$ defined by two coordinate systems $(\alpha_{\bullet}, \check{\alpha}^{\bullet})$ and $(\beta_{\bullet}, \check{\beta}^{\bullet})$, by performing canonical extensions, it suffices to assume that their domain idempotents are equal. Then one can use the same argument as in Lem. 4.4 to show that $(\alpha_{\bullet}, \check{\alpha}^{\bullet})$ and $(\beta_{\bullet}, \check{\beta}^{\bullet})$ define the same ${}^{\psi}\text{Tr}$. Finally, similar to the proof of Prop. 4.5, one shows that ${}^{\psi}\text{Tr}$ is symmetric. $\square$

Example 8.6. Let $M = Ae$ and $B = eAe$ where $e \in A$ is an idempotent. Then the identity map on Ae gives a right coordinate system. From this, one sees that if $\psi \in \text{SLF}(A)$ then

$${}^{\psi}\text{Tr} = \psi|_{eAe}$$

Example 8.7. More generally, let $M = (Ae)^{\oplus n}$ and $B = \text{End}_{A,-}(M)^{\text{op}}$. So $B = eAe \otimes \mathbb{C}^{n \times n}$. Let

$$\text{tr} : \mathbb{C}^{n \times n} \rightarrow \mathbb{C}$$

be the standard trace on $\mathbb{C}^{n \times n}$. A right coordinate system can be chosen to be the n canonical embeddings $Ae \rightarrow (Ae)^{\oplus n}$ and the n canonical projections $(Ae)^{\oplus n} \rightarrow Ae$. Then one easily sees that

$${}^{\psi}\text{Tr} = \psi|_{eAe} \otimes \text{tr}$$

Proposition 8.8. *Assume that M is A -projective. Let $p \in B$ be an idempotent. Let $\psi \in \text{SLF}(A)$. Let ${}^{\psi}\text{Tr}_M : B \rightarrow \mathbb{C}$ be the pseudotrace associated to M . Then the pseudotrace ${}^{\psi}\text{Tr}_{Mp} : pBp \rightarrow \mathbb{C}$ associated to the A -(pBp) bimodule Mp is equal to ${}^{\psi}\text{Tr}_M|_{pBp}$, i.e.*

$${}^{\psi}\text{Tr}_{Mp} = {}^{\psi}\text{Tr}_M|_{pBp}$$

Proof. Let $(\beta_{\bullet}, \check{\beta}^{\bullet})$ be a right coordinate system (with domain idempotent $e \in A$) as in Def. 8.1. Then one has a right coordinate system

$$\begin{aligned} \gamma_j &\in \text{Hom}_{A,-}(Ae, Mp) & \check{\gamma}^j &: \text{Hom}_{A,-}(Mp, Ae) \\ \gamma_j(ae) &= \beta_j(ae)p & \check{\gamma}^j(\xi p) &= \check{\beta}^j(\xi p) \end{aligned}$$

noting that $Mp \leq M$, and hence $\check{\gamma}^j$ is simply the restriction of β^j to Mp . Using (8.2) one computes that for each $y \in B$,

$$\begin{aligned} \psi \text{Tr}_{Mp}(pyp) &= \sum_j \psi(\check{\gamma}^j \circ (pyp)^{\text{op}} \circ \gamma_j(e)) = \sum_j \psi(\check{\beta}^j \circ (pyp)^{\text{op}} \circ \beta_j(e)p) \\ &= \sum_j \psi(\check{\beta}^j \circ (pyp)^{\text{op}} \circ p^{\text{op}} \circ \beta_j(e)) = \sum_j \psi(\check{\beta}^j \circ (pyp)^{\text{op}} \circ \beta_j(e)) = \psi \text{Tr}_M(pyp) \end{aligned}$$

□

9 Equivalence of left and right pseudotraces

Let A, B be algebras where B is unital.

9.1 Preliminary discussion

In this subsection, assume that A is AUF. We shall consider $M \in \text{Coh}_L(A)$ such that the left and the right pseudotrace constructions are both available to the A -($\text{End}_{A,-}(M)^{\text{op}}$) bimodule M . By Rem. 8.2, M needs to be assumed A -projective. One also needs M to be $\text{End}_{A,-}(M)^{\text{op}}$ -projective. In fact, these two conditions are precisely what ensure that both left and right pseudotraces can be defined.

Proposition 9.1. *Let M be an A - B bimodule. Assume that M is A -coherent. Then the following are equivalent.*

- (1) M has a left coordinate system.
- (2) M is B -projective.

Although this proposition will not be used in the current note, we include it here as it may be of use in the future.

Proof. (1) $\Rightarrow$ (2): See Rem. 4.2.

(2) $\Rightarrow$ (1): Let $(e_i)_{i \in \mathcal{I}}$ be as in Def. 5.1. By Rem. 7.2, each $e_i M$ is finite-dimensional. Therefore, the right B -module $e_i M$ has an epimorphism from $B^{\oplus n}$ which splits because M is B -projective (and hence $e_i M$ is projective since $M = \bigoplus_{i \in \mathcal{I}} e_i M$). Therefore, for each $i \in \mathcal{I}$, there is a finite left coordinate system $\alpha_{i,\bullet} \in \text{Hom}_{-,B}(B, e_i M)$ and $\check{\alpha}^{i,\bullet} \in \text{Hom}_{-,B}(e_i M, B)$. Let

$$\begin{aligned} \gamma_{i,\bullet} &\in \text{Hom}_{-,B}(B, M) & \check{\gamma}^{i,\bullet} &\in \text{Hom}_{-,B}(M, B) \\ \gamma_{i,\bullet}(b) &= \alpha_{i,\bullet}(b) & \check{\gamma}^{i,\bullet}(\xi) &= \check{\alpha}^{i,\bullet}(e_i \xi) \end{aligned}$$

Then one checks easily that $(\gamma_{i,\bullet}, \check{\gamma}^{i,\bullet})_{i \in \mathcal{I}}$ is a left coordinate system of M .

□

9.2 Calculation of some left pseudotraces

In this subsection, A is not assumed to be AUF. Let M be an A - B bimodule.

The goal of this subsection is to prepare for the proof of the main Thm. 9.4. The following theorem is dual to Prop. 8.8.

Theorem 9.2. *Assume that M has a left coordinate system. Let $p \in B$ be a generating idempotent. Then the following are true.*

1. *The A -(pBp) bimodule Mp has a left coordinate system.*
2. *Let $\phi \in \text{SLF}(B)$. Then on A , the pseudotraces associated to $\phi|_{pBp}$ and Mp is equal to the pseudotraces associated to ϕ and M . Namely,*

$$\text{Tr}_{Mp}^{\phi|_{pBp}} = \text{Tr}_M^\phi \quad (9.1)$$

In this theorem, we do not require that A is AUF.

Proof. Choose a left coordinate system for M :

$$\alpha_i \in \text{Hom}_{-,B}(B, M) \quad \check{\alpha}^i \in \text{Hom}_{-,B}(M, B) \quad i \in \mathfrak{I}$$

Since p is generating, similar to the proof of Thm. 6.5, we can find finitely many elements u_k, v_k in B such that

$$\begin{aligned} v_k u_k &= p_k & u_k v_k &= q_k \\ u_k &\in q_k B p_k & v_k &\in p_k B q_k \end{aligned}$$

where each $p_k, q_k \in B$ are idempotents, $1_B = \sum_k q_k$ is a primitive orthogonal decomposition of 1_B , and $p_k \leq p$ for each k .³ Let

$$\begin{aligned} \theta_{i,k} &\in \text{Hom}_{-,pBp}(pBp, Mp) & \check{\theta}^{i,k} &\in \text{Hom}_{-,pBp}(Mp, pBp) \\ \theta_{i,k}(pyp) &= \alpha_i(u_k \cdot pyp) & \check{\theta}^{i,k}(\xi p) &= v_k \cdot \check{\alpha}^i(\xi p) \end{aligned}$$

noting that $\alpha_i(u_k \cdot pyp) = \alpha_i(u_k)pyp \in Mp$ and $v_k \cdot \check{\alpha}^i(\xi p) = v_k \cdot \check{\alpha}^i(\xi)p \in p_k B p \subset pBp$.

For each $\xi \in M$, note that if $\check{\alpha}^i(\xi) = 0$, then $\check{\theta}^{i,k}(\xi p) = v_k \check{\alpha}^i(\xi)p = 0$. Therefore, $\check{\theta}^{i,k}(\xi p) = 0$ for all but finitely many i and k . Moreover, we compute

$$\begin{aligned} \sum_{i,k} \theta_{i,k} \circ \check{\theta}^{i,k}(\xi p) &= \sum_{i,k} \theta_{i,k}(v_k \check{\alpha}^i(\xi p)) = \sum_{i,k} \alpha_i(u_k v_k \check{\alpha}^i(\xi p)) \\ &= \sum_{i,k} \alpha_i(q_k \check{\alpha}^i(\xi p)) = \sum_i \alpha_i \circ \check{\alpha}^i(\xi p) = \xi p \end{aligned}$$

where all the sums are finite. This proves that $(\theta, \check{\theta})$ satisfies Def. 4.1-(a). It is easy to check Def. 4.1-(b). So we have proved that $(\theta, \check{\theta})$ is a left coordinate system of Mp .

³So p_k, q_k are similar to ε_{k_i}, e_i in the proof of Thm. 6.5.

It remains to check (9.1). Choose any $x \in A$. By (4.3) and the fact that $1_{pBp} = p$,

$$\begin{aligned} \mathrm{Tr}_{Mp}^{\phi|_{pBp}}(x) &= \sum_{i,k} \phi(\check{\theta}^{i,k} \circ x \circ \theta_{i,k}(p)) = \sum_{i,k} \phi(\check{\theta}^{i,k} \circ x \circ \alpha_i(u_k p)) \\ &= \sum_{i,k} \phi(\check{\theta}^{i,k} \circ x \circ \alpha_i(u_k)) = \sum_{i,k} \phi(v_k \cdot \check{\alpha}^i(x \circ \alpha_i(u_k))) \end{aligned}$$

Since $\check{\alpha}^i, x, \alpha_i$ commute with the right multiplication by v_k , and since ϕ is symmetric,

$$\begin{aligned} \mathrm{Tr}_{Mp}^{\phi|_{pBp}}(x) &= \sum_{i,k} \phi(\check{\alpha}^i(x \circ \alpha_i(u_k))v_k) = \sum_{i,k} \phi(\check{\alpha}^i(x \circ \alpha_i(u_k v_k))) \\ &= \sum_i \phi(\check{\alpha}^i(x \circ \alpha_i(1_B))) = \mathrm{Tr}_M^\phi(x) \end{aligned}$$

This finishes the proof of (9.1). $\square$

Corollary 9.3. *Assume that M has a left coordinate system. Let $n \in \mathbb{Z}_+$. Let $\tilde{B} = B \otimes \mathbb{C}^{n \times n}$. Then the A - $\tilde{B}$ bimodule $M^{\oplus n} \simeq M \otimes \mathbb{C}^{1,n}$ has a left coordinate system. Moreover, for each $\phi \in \mathrm{SLF}(B)$, we have*

$$\mathrm{Tr}_{M^{\oplus n}}^{\phi \otimes \mathrm{tr}} = \mathrm{Tr}_M^\phi \quad (9.2)$$

as pseudotraces on A associated to $\phi \otimes \mathrm{tr} \in \mathrm{SLF}(\tilde{B})$ and $\phi \in \mathrm{SLF}(B)$, respectively.

Recall that $\mathrm{tr} \in \mathrm{SLF}(\mathbb{C}^{n \times n})$ is the standard trace on the $n \times n$ matrix algebra.

Proof. Choose a left coordinate system

$$\alpha_i \in \mathrm{Hom}_{-,B}(B, M) \quad \check{\alpha}^i \in \mathrm{Hom}_{-,B}(M, B)$$

of M . Define

$$\gamma_i \in \mathrm{Hom}_{-,\tilde{B}}(\tilde{B}, M^{\oplus n}) \quad \check{\gamma}^i \in \mathrm{Hom}_{-,\tilde{B}}(M^{\oplus n}, \tilde{B})$$

such that

$$\begin{aligned} \gamma_i \begin{bmatrix} y_{1,1} & \cdots & y_{1,n} \\ \vdots & & \\ y_{n,1} & \cdots & y_{n,n} \end{bmatrix} &= [\alpha_i(1_B), 0, \dots, 0] \begin{bmatrix} y_{1,1} & \cdots & y_{1,n} \\ \vdots & & \\ y_{n,1} & \cdots & y_{n,n} \end{bmatrix} = [\alpha_i(y_{1,1}), \dots, \alpha_i(y_{1,n})] \\ \check{\gamma}^i[\xi_1, \dots, \xi_n] &= \begin{bmatrix} \check{\alpha}^i(\xi_1) & \cdots & \check{\alpha}^i(\xi_n) \\ 0 & \cdots & 0 \\ \vdots & \vdots & \vdots \\ 0 & \cdots & 0 \end{bmatrix} \end{aligned}$$

One checks easily that this is a left coordinate system of $M^{\oplus n}$. Now (9.2) follows by applying Thm. 9.2 to the A - $\tilde{B}$ bimodule $M^{\oplus n}$ and the generating projection $p \in \tilde{B}$, where p is the matrix whose $(1, 1)$ -entry is 1 and other entries are 0. $\square$

9.3 The main theorem

Assume that A is strongly AUF (cf. Cor. 6.4) so that A has a projective generator (cf. Prop. 7.8). The following generalization of Thm. 6.6 is the main theorem of this note.

Theorem 9.4. *Assume that $M \in \text{Coh}_L(A)$ is a projective generator. Assume that $B = \text{End}_{A,-}(M)^{\text{op}}$ so that M is an A - B bimodule. Then M has left and right coordinate systems. Moreover, we have a linear isomorphism*

$$\text{SLF}(A) \xrightarrow{\sim} \text{SLF}(B) \quad \psi \mapsto \psi \text{Tr} \quad (9.3a)$$

whose inverse map is

$$\text{SLF}(B) \xrightarrow{\sim} \text{SLF}(A) \quad \phi \mapsto \text{Tr}^\phi \quad (9.3b)$$

Of course, both pseudotraces are associated to M ; we have suppressed the subscript M .

Proof. Note that $\dim B < +\infty$ by Prop. 7.9. So $\dim \text{SLF}(B) < +\infty$. Since $M \in \text{Coh}_L(A)$ is A -projective, by Rem. 8.2, M has a right coordinate system. By Cor. 7.12, we may assume that $M = G \cdot p$ where

- $G = (Ae)^{\oplus n}$ for some $n \in \mathbb{Z}_+$ and generating idempotent $e \in A$.
- $M = Gp$ where p is a generating idempotent of $\tilde{B} = \text{End}_{A,-}(G)^{\text{op}} = eAe \otimes \mathbb{C}^{n \times n}$.
- $B = p\tilde{B}p$ (by Prop. 7.10).

By Thm. 6.5 and Cor. 9.3, G has a left coordinate system. Therefore, by Thm. 9.2, M has a left coordinate system.

By Thm. 6.6, we have $\dim \text{SLF}(A) = \dim \text{SLF}(eAe)$. Clearly we have a linear isomorphism

$$\text{SLF}(eAe) \xrightarrow{\sim} \text{SLF}(eAe \otimes \mathbb{C}^{n \times n}) \quad \omega \mapsto \omega \otimes \text{tr}$$

So $\dim \text{SLF}(eAe) = \dim \text{SLF}(\tilde{B})$. By Thm. 6.6, we have $\dim \text{SLF}(\tilde{B}) = \dim \text{SLF}(B)$. This proves $\dim \text{SLF}(A) = \dim \text{SLF}(B) < +\infty$.

Choose any $\psi \in \text{SLF}(A)$. By Exp. 8.7, ${}^\psi \text{Tr}_G : \tilde{B} \rightarrow \mathbb{C}$ equals $\psi|_{eAe} \otimes \text{tr}$. By Prop. 8.8, on $B = p(eAe \otimes \mathbb{C}^{n \times n})p$ we have

$${}^\psi \text{Tr}_M = (\psi|_{eAe} \otimes \text{tr})|_B =: \phi$$

Now $\phi \in \text{SLF}(B)$. By Thm. 9.2 and Cor. 9.3,

$$\text{Tr}_M^\phi = \text{Tr}_G^{(\psi|_{eAe} \otimes \text{tr})|_B} = \text{Tr}_G^{\psi|_{eAe} \otimes \text{tr}} = \text{Tr}_{Ae}^{\psi|_{eAe}}$$

By Thm. 6.6, $\text{Tr}_{Ae}^{\psi|_{eAe}} = \psi$. So $\text{Tr}_M^\phi = \psi$. We have thus proved that $(9.3b) \circ (9.3a)$ is the identity map on $\text{SLF}(A)$. This finishes the proof. $\square$

10 Equivalence of non-degeneracy of left and right pseudotraces

Definition 10.1. Let A be an algebra and $\psi \in \text{SLF}(A)$. We say that ψ is **non-degenerate** if

$$\{x \in A : \psi(xA) = 0\} \equiv \{x \in A : \psi(xy) = 0, \forall y \in A\}$$

is zero.

In the following, A is always assumed to be AUF.

Lemma 10.2. *Let $e \in A$ be an idempotent, and let $\psi \in \text{SLF}(A)$. If ψ is non-degenerate, then the restriction $\psi|_{eAe}$ is non-degenerate. Conversely, if $\psi|_{eAe}$ is non-degenerate and e is generating, then ψ is non-degenerate.*

Proof. Assume that ψ is non-degenerate. Choose $x \in eAe$ such that $\psi(xeAe) = 0$. Then

$$\psi(xA) = \psi(exeA) = \psi(xeAe) = 0$$

and hence $x = 0$. Therefore $\psi|_{eAe}$ is non-degenerate.

Conversely, assume that $\psi|_{eAe}$ is non-degenerate and e is generating. Choose $x \in A$ such that $\psi(xA) = 0$. Then for each $a, b \in A$,

$$\psi(eaxbe \cdot eAe) = \psi(eaxbeAe) = \psi(xbeAea) = 0$$

Therefore $eaxbe = 0$. Since b is arbitrary, we have $eaxAe = 0$. Since e is generating, it is not hard to show that the left A -module Ae is faithful. (See for example Lem. 11.6.) It follows from that $eax = 0$. Therefore $eAx = 0$. Similarly, eA is a faithful right A -module. Hence $x = 0$. This proves the non-degeneracy of ψ . $\square$

Proposition 10.3. *Assume that $\psi \in \text{SLF}(A)$ is non-degenerate. Let $M \in \text{Coh}_L(A)$ be projective, and let $B = \text{End}_{A,-}^0(M)$. Then the right pseudotrace ${}^\psi\text{Tr} \in \text{SLF}(B)$ is non-degenerate.*

Proof. By Prop. 2.6, M can be viewed as a direct summand of $\bigoplus_{i=1}^n Ae_i$ where each $e_i \in A$ is an idempotent. Let $e \in A$ be an idempotent such that $e \geq e_i$ for all i . Then M is a direct summand of $(Ae)^{\oplus n}$. By Prop. 1.2, we have $\text{End}_{A,-}^0(Ae)^{\text{op}} = eAe$, and hence $\text{End}_{A,-}^0((Ae)^{\oplus n}) = eAe \otimes \mathbb{C}^{n \times n}$. By Cor. 1.4, there is an idempotent $p \in eAe \otimes \mathbb{C}^{n \times n}$ such that $M = (Ae)^{\oplus n}p$. By Lem. 10.2, $\psi|_{eAe}$ is non-degenerate, and hence $\psi|_{eAe} \otimes \text{tr} : eAe \otimes \mathbb{C}^{n \times n} \rightarrow \mathbb{C}$ is non-degenerate. By Lem. 10.2 again, the restriction of $\psi|_{eAe} \otimes \text{tr}$ to $p(eAe \otimes \mathbb{C}^{n \times n})p$ (which is B due to Prop. 7.10) is non-degenerate. But this restriction is exactly ${}^\psi\text{Tr}$ due to Exp. 8.7 and Prop. 8.8. $\square$

Theorem 10.4. *Assume that A is strongly AUF. Then in Thm. 9.4, for any $\psi \in \text{SLF}(A)$, the non-degeneracy of ψ and of ${}^\psi\text{Tr}$ are equivalent.*

Proof. We use the notation in the proof of Thm. 9.4. From that proof, we know ${}^\psi\text{Tr} = (\psi|_{eAe} \otimes \text{tr})|_B$. By Lem. 10.2, ψ is non-degenerate iff $\psi|_{eAe}$ is so, and $\psi|_{eAe} \otimes \text{tr}$ is non-degenerate iff $(\psi|_{eAe} \otimes \text{tr})|_B$ is so. The equivalence of the non-degeneracy of $\psi|_{eAe}$ and of $\psi|_{eAe} \otimes \text{tr}$ is obvious. The proof is finished. $\square$

11 Classification of strongly AUF algebras

In this section, we fix an AUF algebra A .

Definition 11.1. For each left A -module M , let M^* be the space of linear functionals, which has a right A -module structure defined by

$$(\phi a)(m) = \phi(am) \quad \text{for all } a \in A, m \in M$$

We define the **quasicoherent dual**

$$\begin{aligned} M^\vee &= \{\phi \in M^* : \phi \in \phi \cdot A\} \\ &= \{\phi \in M^* : \text{there exists an idempotent } e \in A \text{ such that } \phi = \phi e\} \end{aligned}$$

By Def. 2.2, $M^\vee$ is the largest right A -submodule of M that is quasicoherent.

Remark 11.2. Let $M \in \text{QCoh}_L(A)$. Let $(e_i)_{i \in \mathfrak{I}}$ be as in Def. 5.1. Then, as vector spaces, we clearly have

$$M = \bigoplus_{i \in \mathfrak{I}} e_i M \quad M^* = \prod_{i \in \mathfrak{I}} (e_i M)^*$$

It follows easily that

$$M^\vee = \bigoplus_{i \in \mathfrak{I}} (e_i M)^*$$

Definition 11.3. For each $M \in \text{QCoh}_L(A)$, we let

$$\text{End}^0(M) = M \otimes_{\mathbb{C}} M^\vee$$

viewed as a subalgebra of $\text{End}(M)$.⁴ Suppose that B is an algebra, and M has a right B -module structure commuting with the left action of A , we let

$$\text{End}_{-,B}^0(M) = \{T \in \text{End}^0(M) : (T\xi)b = T(\xi b) \text{ for all } \xi \in M, b \in B\} \quad (11.1)$$

Remark 11.4. Let $M \in \text{Coh}_L(A)$. By Rem. 7.2 we have $\dim e_i M < +\infty$. It follows from Rem. 11.2 that

$$\text{End}^0(M) = \{T \in \text{End}(M) : Te_i = 0 \text{ for all but finitely many } i \in \mathfrak{I}\}$$

Proposition 11.5. Choose $M \in \text{Coh}_L(A)$, and let $B = \text{End}_{A,-}(M)^{\text{op}}$. Then for each generating idempotent $p \in B$, we have a linear isomorphism

$$\text{End}_{-,B}^0(M) \xrightarrow{\cong} \text{End}_{-,pBp}^0(Mp) \quad S \mapsto S|_{Mp} \quad (11.2)$$

⁴That is, for each $\xi \in M$, $\phi \in M^\vee$, the operator $\xi \otimes \phi$ sends each $\eta \in M$ to $\phi(\eta) \cdot \xi$.

Proof. Step 1. Let $\hat{B} = B^{\text{op}} = \text{End}_{A,-}(M)$, and let $\hat{p} \in \hat{B}$ be the opposite element of p . Then M has a left $\hat{B}$ -module structure commuting with the left action of A , and R_p is the left multiplication by $\hat{p}$.

For each $S \in \text{End}_{-,B}^0(M)$, note that $S|_{Mp} = S|_{\hat{p}M}$ maps $\hat{p}M$ into $\hat{p}M$, because $S\hat{p}\xi = \hat{p}S\xi \in \hat{p}M$ for each $\xi \in M$. It is clear that $S|_{Mp}$ commutes with the action of $\hat{p}\hat{B}\hat{p}$. That $S|_{Mp}$ belongs to $\text{End}^0(M)$ can be checked from Rem. 11.4. This proves that $S|_{Mp}$ belongs to $\text{End}_{-,pBp}^0(Mp)$. We have thus proved that the linear map (11.2) is well-defined.

Step 2. Let us prove the surjectivity of (11.2). By Rem. 5.4, B is finite-dimensional. Therefore, we have an orthogonal primitive decomposition $1_{\hat{B}} - \hat{p} = f_1 + \cdots + f_n$ in $\hat{B}$. In this case, we have

$$M = \hat{p}M \oplus f_1M \oplus \cdots \oplus f_nM$$

By Prop. 6.2, for each $1 \leq i \leq n$, f_i is isomorphic to a sub-idempotent q_i of $\hat{p}$, i.e., there exist $u_i \in f_i\hat{B}q_i$ and $v_i \in q_i\hat{B}f_i$ such that $u_iv_i = f_i$ and $v_iu_i = q_i \leq \hat{p}$ (where $q_i \in \hat{B}$ is an idempotent).

Now, we choose $T \in \text{End}_{-,pBp}^0(Mp) = \text{End}_{-,pBp}^0(\hat{p}M)$. Define a linear map

$$S : M \rightarrow M \quad \xi \mapsto T(\hat{p}\xi) + \sum_{i=1}^n u_i T(v_i \xi) \quad (11.3)$$

By Rem. 11.4, we have $S \in \text{End}^0(M)$. We claim that S commutes with the action of $\hat{B}$ (and hence $S \in \text{End}_{-,B}^0(M)$). If this is proved, then since T clearly equals $S|_{Mp} = S|_{\hat{p}M}$ (because $v_i\hat{p} = 0$, see below), the proof of the surjectivity of (11.2) is complete.

Note that since $\hat{p}, f_1, \dots, f_n$ are mutually orthogonal, we have

$$\begin{aligned} u_i u_j &= 0 & v_i v_j &= 0 & \text{for all } i, j \\ v_j u_i &= 0 & & \text{for all } i \neq j \\ v_i \hat{p} &= 0 & \hat{p} u_i &= 0 & \text{for all } i \end{aligned}$$

Using this observation and the fact that $T : \hat{p}M \rightarrow \hat{p}M$ commutes the left action of $\hat{p}\hat{B}\hat{p}$, we compute that for each j and $\xi \in M$,

$$\begin{aligned} S(v_j \xi) &= T(\hat{p}v_j \xi) + 0 = T(v_j \xi) \\ v_j S(\xi) &= v_j T(\hat{p}\xi) + v_j u_j T(v_j \xi) \stackrel{v_j u_j = q_j \in \hat{p}\hat{B}\hat{p}}{=} 0 + T(q_j v_j \xi) = T(v_j \xi) \end{aligned}$$

and hence $S(v_j \xi) = v_j S(\xi)$; similarly,

$$\begin{aligned} S(u_j \xi) &= T(\hat{p}u_j \xi) + u_j T(v_j u_j \xi) \stackrel{v_j u_j = q_j \in \hat{p}\hat{B}\hat{p}}{=} 0 + u_j q_j T(\hat{p}\xi) = u_j T(\hat{p}\xi) \\ u_j S(\xi) &= u_j T(\hat{p}\xi) + 0 = u_j T(\hat{p}\xi) \end{aligned}$$

and hence $S(u_j \xi) = u_j S(\xi)$. Moreover, for each $b \in \hat{B}$ we have

$$S(\hat{p}b\hat{p}\xi) = T(\hat{p}b\hat{p}\xi) + 0 = \hat{p}b\hat{p}T(\hat{p}\xi)$$

$$\hat{p}b\hat{p}S(\xi) = \hat{p}b\hat{p}T(\hat{p}\xi) + 0 = \hat{p}b\hat{p}T(\hat{p}\xi)$$

and hence $S(\hat{p}b\hat{p}\xi) = \hat{p}b\hat{p}S(\xi)$. This proves that S commutes with the left action of $\hat{B}$, since $\hat{B}$ is generated by $\{u_i, v_i : 1 \leq i \leq n\}$ and $\hat{p}\hat{B}\hat{p}$ —to see this, note that for each $b \in \hat{B}$, by setting $f_0 = u_0 = v_0 = \hat{p}$, we have

$$b = \sum_{i,j=0}^n f_i b f_j = \sum_{i,j=0}^n u_i b_{i,j} v_j$$

where each $b_{i,j} := v_i b u_j$ commutes with the left actions of A and satisfies $b_{i,j} = \hat{p}b_{i,j}\hat{p}$, and hence belongs to $\hat{p}\hat{B}\hat{p}$.

Step 3. If $S \in \text{End}_{-,B}^0(M)$ and $S|_{\hat{p}M} = 0$, then for each $\xi \in M$, we have

$$S(\xi) = S(\hat{p}\xi) + \sum_{i=1}^n S(f_i\xi) = S(\hat{p}\xi) + \sum_{i=1}^n u_i S(v_i\xi)$$

where $\hat{p}\xi, v_j\xi \in \hat{p}M$. Therefore $S = 0$. This proves that (11.2) is injective. $\square$

Lemma 11.6. *Suppose that $e \in A$ is a generating idempotent. Then we have a linear isomorphism*

$$A \xrightarrow{\cong} \text{End}_{-,eAe}^0(Ae) \quad (11.4)$$

sending each $a \in A$ to the left multiplication by a .

Proof. It is obvious that the left action on Ae by $a \in A$ belongs to $\text{End}_{-,eAe}^0(Ae)$. Therefore, the map (11.4) is well-defined.

Suppose that the left multiplication of $a \in A$ on Ae is zero. Then $aAe = 0$. Since A is AUF and hence almost unital, there is an idempotent $p \in A$ such that $a = ap$. Since e is generating, by Cor. 7.7, Ae is a generator of $\text{Coh}_L(A)$. Therefore, Ap is a quotient module of $(Ae)^{\oplus n}$ for some $n \in \mathbb{Z}_+$. Thus aAp is a quotient space of $(aAe)^{\oplus n}$, and hence $aAp = 0$. This proves $ap = 0$, and hence $a = 0$. We have thus proved that (11.4) is injective.

Choose $T \in \text{End}_{-,eAe}^0(Ae)$. Since $T \in \text{End}^0(Ae)$, by Rem. 11.2, there is an idempotent $f \in A$ such that $T = fTf$. It follows that $fTf|_{fAe}$ belongs to $\text{End}_{-,eAe}(fAe)$. Since A is AUF, we may enlarge f so that $e \leq f$ also holds. We claim that $\text{End}_{-,eAe}(fAe)$ consists of the left multiplications by elements of fAf . If this is true, then $T|_{fAe} = fTf|_{fAe}$ is the left multiplication by faf for some $a \in A$. It follows that for any $b \in A$, we have $Tbe = Tfbe = fafbe$, and hence T is the left multiplication by faf on Ae , finishing the proof that (11.4) is surjective.

By Cor. 6.3, the idempotent $e \in fAf$ is generating in fAf . Applying Prop. 11.5 to the finite-dimensional unital algebra fAf and its (finite-dimensional) coherent left module fAf , we see that $\text{End}_{-,eAe}(fAe) = fAf|_{fAe}$. This proves the claim. $\square$

Theorem 11.7. *Suppose that A is strongly AUF, and let G be a projective generator of $\text{Coh}_L(A)$ (which exists due to Prop. 7.8). Set $B = \text{End}_{A,-}(G)^{\text{op}}$. Regard G as an A - B bimodule. Then we have a linear isomorphism*

$$A \xrightarrow{\cong} \text{End}_{-,B}^0(G) \quad (11.5)$$

sending each $a \in A$ to the left multiplication of a on G .

Proof. By Cor. 6.4 and Prop. 7.8, A has a generating idempotent e . If $G = Ae$, then $\text{End}_{A,-}(G) = eAe$ due to Prop. 1.2. Therefore, by Lem. 11.6, the map (11.5) is bijective.

If $G = (Ae)^{\oplus n}$ where $n \in \mathbb{Z}_+$, one easily checks that $B = eAe \otimes \mathbb{C}^{n \times n}$ where $\mathbb{C}^{n \times n}$ is the matrix algebra of order n . The bijectivity of (11.5) then follows easily.

Finally, let G be any general projective generator. By Cor. 7.12, we may assume that $G = (Ae)^{\oplus n}p$ where $n \in \mathbb{Z}_+$, and p is a generating idempotent of $\tilde{B} = \text{End}_{A,-}((Ae)^{\oplus n})^{\text{op}} \simeq eAe \otimes \mathbb{C}^{n \times n}$. By Prop. 7.10, we have $B = p\tilde{B}p$. Therefore, by Prop. 11.5, the map

$$\text{End}_{-, \tilde{B}}^0((Ae)^{\oplus n}) \rightarrow \text{End}_{-, B}^0(G)$$

sending each S to $S|_G$ is bijective. By the previous paragraph, the map

$$A \rightarrow \text{End}_{-, \tilde{B}}^0((Ae)^{\oplus n})$$

sending each a to the left multiplication by a is bijective. Therefore, their composition, namely (11.5), is bijective. $\square$

Remark 11.8. In Thm. 11.7, the right B -module G is a **projective generator** in the category $\text{Mod}^R(B)$ of right B -modules—that is, G is projective in $\text{Mod}^R(B)$, and any object in $\text{Mod}^R(B)$ has an epimorphism from a (possibly infinite) direct sum of G .

Proof. The projectivity of G in $\text{Mod}^R(B)$ is due to Thm. 9.4 and Rem. 4.2. Using the notation in the proof of Thm. 11.7, we may assume $G = (Ae)^{\oplus n}p$ and $B = p(eAe \otimes \mathbb{C}^{n \times n})p$ where $e \in A$ and $p \in eAe \otimes \mathbb{C}^{n \times n}$ are generating idempotents. Since B is unital, B is generating in $\text{Mod}^R(B)$. Therefore $(eAe \otimes \mathbb{C}^{n \times n})p$ is generating in $\text{Mod}^R(B)$. Since $(eAe \otimes \mathbb{C}^{n \times n})p$ is a direct sum of $(eAe \otimes \mathbb{C}^{1 \times n}) = (eAe)^{\oplus n}p = eG$, we conclude that eG is generating in $\text{Mod}^R(B)$. Therefore G is generating in $\text{Mod}^R(B)$. $\square$

Theorem 11.9. *Let A be an algebra. The following are equivalent.*

- (1) A is strongly AUF.
- (2) A is isomorphic to $\text{End}_{-, B}^0(M)$ where B is a unital finite-dimensional algebra, M is a projective generator in $\text{Mod}^R(B)$, the vector space M has a grading

$$M = \bigoplus_{i \in \mathcal{I}} M(i)$$

where each $M(i)$ is finite-dimensional and is preserved by the right action of B , and $\text{End}_{-, B}^0(M)$ is defined by

$$\begin{aligned} \text{End}_{-, B}^0(M) &:= \{T \in \text{End}(M) : (Tm)b = T(mb) \text{ for all } m \in M, b \in B, \\ &\quad T|_{M(i)} = 0 \text{ for all but finitely many } i \in \mathcal{I}\} \end{aligned}$$

Proof. The direction (1) $\Rightarrow$ (2) follows from Thm. 11.7 and Rem. 11.8. Let us prove the other direction.

Assume that $A = \text{End}_{-, B}^0(M)$ where $\text{End}_{-, B}^0(M)$ is described as in (2). Let e_i be the projection of M onto $M(i)$. Then e_i clearly belongs to A , and each $T \in A$ can be written as $T = \sum_{i, j \in \mathcal{I}} e_i T e_j$ where $e_i T e_j = 0$ for all but finitely many i, j . This proves that A is AUF.

Since M is a projective generator in $\text{Mod}^R(B)$, for each finite subset $I \subset \mathfrak{I}$, $M_I := \bigoplus_{i \in I} M(i)$ is projective in $\text{Mod}^R(B)$ (since it is a direct summand of M). Let $1_B = p_1 + \cdots + p_n$ be an orthogonal primitive decomposition of 1_B in B . By Thm. 5.5, irreducible finite-dimensional right B -modules are precisely those that are isomorphic to $p_k B / \text{rad}(p_k B)$ for some k . Since M is generating in $\text{Mod}^R(B)$, it has an epimorphism to $p_k B / \text{rad}(p_k B)$ for each k . This epimorphism must restrict to a nonzero morphism (and hence an epimorphism) $M(i_k) \rightarrow p_k B / \text{rad}(p_k B)$. Let $I = \{i_1, \dots, i_n\}$. Then M_I has an epimorphism to each irreducible right B -module. It follows from Prop. 7.6 that M_I is a projective generator in the category of finite-dimensional right B -modules.

Let $e_I = \sum_{i \in I} e_i$, which is an idempotent in $\mathcal{A}$. We claim that e_I is a generating idempotent in $\mathcal{A}$, which will complete the proof that $\mathcal{A}$ is strongly AUF.

Let ε be any primitive idempotent of $\mathcal{A}$. Then εM is a finite-dimensional right B -module, since any element of $\mathcal{A}$ has finite range when acting on M . Moreover, since ε is primitive in $\mathcal{A}$, the right B -module εM is indecomposable. Since εM is a direct summand of the projective right B -module M , it follows that εM is a finite-dimensional indecomposable projective right B -module. Therefore, since $M_I = e_I M$ is a projective generator, similar to the the end of the proof of Thm. 7.11, we conclude that the right B -module εM is isomorphic to a direct summand of $e_I M$. Thus, by Thm. 7.10, ε is isomorphic to a subidempotent of e_I in $\mathcal{A}$. This proves the claim that e_I is generating. $\square$

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